C1 Assignment : Integral Calculus

SINGLE CORRECT TYPE QUESTIONS

I04/1/001. Area bounded by $x^2y^2 + y^4 - x^2 - 5y^2 + 4 = 0$ is equal to :

(A) $\frac{4\pi}{3}+\sqrt{2}$ (B) $\frac{4\pi}{3}-\sqrt{2}$ (C) $\frac{4\pi}{3}+2\sqrt{3}$ (D) none of these

$x^2y^2 + y^4 - x^2 - 5y^2 + 4 = 0$ से परिबद्ध क्षेत्रफल होगा :

(A) $\frac{4\pi}{3}+\sqrt{2}$ (B) $\frac{4\pi}{3}-\sqrt{2}$ (C) $\frac{4\pi}{3}+2\sqrt{3}$ (D) इनमें से कोई नहीं

Ans. C

Sol. $x^2y^2 + y^4 - x^2 - 5y^2 + 4 = 0$

$(x^2 + y^2 - 4)(y^2 - 1) = 0$

I04/1/002. Area bounded by $y = -x^2 + 6x - 5$, $y = -x^2 + 4x - 3$ and $y = 3x - 15$, for $x > 1$, is :

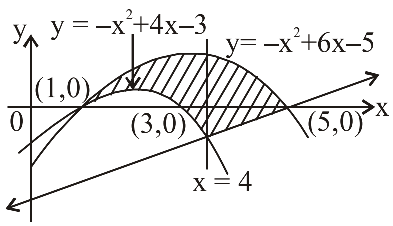
(A) 73 (B) $\frac{13}{6}$ (C) $\frac{73}{6}$ (D) None of these

$x > 1$ के लिए $y = -x^2 + 6x - 5$, $y = -x^2 + 4x - 3$ तथा $y = 3x - 15$ से परिबद्ध क्षेत्रफल होगा :

(A) 73 (B) $\frac{13}{6}$ (C) $\frac{73}{6}$ (D) इनमें से कोई नहीं

Ans. C

Sol.



$$A = \int_1^4 \left\{\left(-x^2+6x-5\right)-\left(-x^2+4x-3\right)\right\}dx + \int_4^5 \left(-x^2+6x-5\right)-\left(3x-15\right)dx = \frac{73}{6}$$

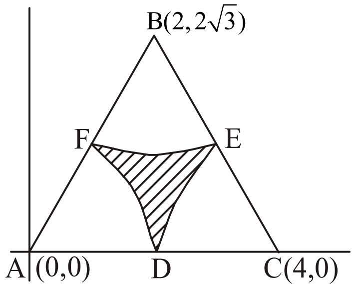
I04/1/003. A point P moves inside a triangle formed by A(0,0), $B\left(2,2\sqrt{3}\right)$, C(4, 0) such that {min(PA, PB, PC)} = 2, then the area bounded by the curve traced by P is :

A(0,0), $B\left(2,2\sqrt{3}\right)$, C(4, 0) से निर्मित एक त्रिभुज में एक बिन्दु P इस प्रकार गति करता है कि {min(PA, PB, PC)} = 2, तब P द्वारा बनाये गये वक्र से परिबद्ध क्षेत्रफल होगा :

(A) $3\sqrt{3}-\frac{3\pi}{2}$ (B) $4\sqrt{3}-2\pi$ (C) $\sqrt{3}-\frac{\pi}{2}$ (D) 2π

Ans. D

Sol. D, E, F are the mid points of the sides AC, BC, AB respectively.



$$\text{Required area} = \left(\frac{\sqrt{3}}{4}\times 16-3\times\frac{2\pi}{3}\right)=4\sqrt{3}-2\pi$$

I02/1/004. Let f(x) is differentiable function symmetric about x = 2, then the value of $\int_0^4 \cos(\pi x) f'(x)\,dx$ is equal to :

माना f(x) एक अवकलनीय फलन x = 2 के सापेक्ष सममित हो, तब $\int_0^4 \cos(\pi x) f'(x)\,dx$ का मान बराबर होगा :

(A) 3 (B) 2 (C) 1 (D) 0

Ans. D

Sol. Given f(x) = f(4 – x)

$$\Rightarrow I=\int_0^4 \cos(\pi x) f'(x)\,dx=\int_0^4 \cos(4\pi-\pi x) f'(4-x)\,dx=-I\Rightarrow 2I=0$$

I02/1/005. Let $p(x) = a_1x + a_2x^2 + a_3x^3 \ldots\ldots a_{100}x^{100}$, where $a_1 = 1$ and $a_i \in R\ \forall\ i = 2,3,\ldots,100$ then

$\lim\limits_{x\to 0}\left[\frac{1}{x^4}\int_0^x \left\{x^2\left(\sqrt[100]{p(t)+1}\right)-x^2\right\}dt\right]$ is :

(A) $\frac{1}{200}$ (B) $\frac{1}{50}$ (C) 150 (D) None of these

माना $p(x) = a_1x + a_2x^2 + a_3x^3 \ldots\ldots a_{100}x^{100}$, जहाँ $a_1 = 1$ तथा $a_i \in R\ \forall\ i = 2,3,\ldots,100$ तब

$\lim\limits_{x\to 0}\left[\frac{1}{x^4}\int_0^x \left\{x^2\left(\sqrt[100]{p(t)+1}\right)-x^2\right\}dt\right]$ होगा :

(A) $\frac{1}{200}$ (B) $\frac{1}{50}$ (C) 150 (D) इनमें से कोई नहीं

Ans. A

Sol.

$$L=\lim_{x\to 0}\frac{x^2\int_0^x\left(\sqrt[100]{p(t)+1}-1\right)dt}{x^4}$$

$$L=\lim_{x\to 0}\frac{\sqrt[100]{p(t)+1}-1}{2x}\ \text{(Using Libinitz Rule)}$$

$$\Rightarrow L=\frac{1}{200}$$

I02/1/006. Let $\lim_{n\to\infty}\left(\frac{(2.1+n)}{1^2+n.1+n^2}+\frac{(2.2+n)}{2^2+n.2+n^2}+\frac{(2.3+n)}{3^2+n.3+n^2}+....+\frac{(2.n+n)}{3n^2}\right)$ then value of e^L is :

माना $\lim_{n\to\infty}\left(\frac{(2.1+n)}{1^2+n.1+n^2}+\frac{(2.2+n)}{2^2+n.2+n^2}+\frac{(2.3+n)}{3^2+n.3+n^2}+....+\frac{(2.n+n)}{3n^2}\right)$ है तब e^L का मान होगा :

(A) 2 (B) 3 (C) 4 (D) $\frac{3}{2}$

Ans. B

Sol.

$$L=\lim_{n\to\infty}\frac{2.r+n}{r^2+n.r+n^2}=\int_0^1\frac{(2x+1)dx}{x^2+x+1}=\ell n\left(x^2+x+1\right)\Big|_0^1\Rightarrow L=\ell n3$$

I02/1/007. The value of the definite integral $\int_0^{\frac{\pi}{2}}\left(\frac{1+\sin 3x}{1+2\sin x}\right)dx$ equals to :

निश्चित समाकलन $\int_0^{\frac{\pi}{2}}\left(\frac{1+\sin 3x}{1+2\sin x}\right)dx$ का मान बराबर होगा :

(A) $\frac{\pi}{2}$ (B) 1 (C) $\frac{1}{2}$ (D) $\frac{\pi}{4}$

Ans. B

Sol. $I=\int_0^{\frac{\pi}{2}}\left(\frac{1+\sin 3x}{1+2\sin x}\right)dx=\int_0^{\frac{\pi}{2}}\frac{1+3\sin x-4\sin^3 x}{1+2\sin x}dx$

$$=\int_0^{\frac{\pi}{2}}\frac{(1+2\sin x)\left(-2\sin^2 x+\sin x+1\right)}{1+2\sin x}dx=-2\left(\frac{1}{2}\frac{\pi}{2}\right)+1+\frac{\pi}{2}=1$$

I02/1/008. $Q=\int_1^{\infty}\frac{(x-1)^4 dx}{x^8+x^{10}}$ is :

$Q=\int_1^{\infty}\frac{(x-1)^4 dx}{x^8+x^{10}}$ होगा :

(A) $\frac{22}{7}-\pi$ (B) $\pi-\frac{22}{7}$ (C) 0 (D) $2\pi-\frac{22}{7}$

Ans. A

Sol. Let $I=\int_1^{\infty}\frac{(x-1)^4}{x^8+x^{10}}dx$

Let $\frac{1}{x}=y\Rightarrow dx=\frac{-dy}{y^2}$

Now; $I=\int_1^{0}\frac{(1-y)^4(-dy)}{y^4\cdot y^2\left(\frac{1}{y^8}+\frac{1}{y^{10}}\right)}$

$$=\int_0^1\frac{y^4(1-y)^4}{1+y^2}=\int_0^1\left(y^6-4y^5+5y^4-4y^2+4\right)dy-4\int_0^1\frac{dy}{1+y^2}=\frac{22}{7}-\pi$$

I02/1/009. The value of $\left[1+\frac{1}{\sqrt{3}}+\frac{1}{\sqrt{5}}+....+\frac{1}{\sqrt{121}}\right]$ is equal to :

(where [.] denotes greatest integer function)

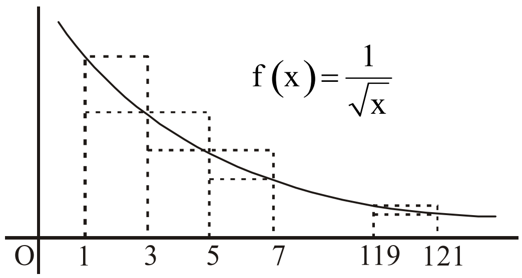
$\left[1+\frac{1}{\sqrt{3}}+\frac{1}{\sqrt{5}}+....+\frac{1}{\sqrt{121}}\right]$ का मान होगा :

(जहाँ [.] महत्तम पूर्णांक फलन को दर्शाता है)

(A) 10 (B) 11 (C) 9 (D) 12

Ans. A

Sol.



Let $S=1+\frac{1}{\sqrt{3}}+\frac{1}{\sqrt{5}}+....+\frac{1}{\sqrt{121}}$

Clearly $2\left(\frac{1}{\sqrt{3}}+\frac{1}{\sqrt{5}}+....+\frac{1}{\sqrt{121}}\right)<\int_{1}^{121}\frac{dx}{\sqrt{x}}<2\left(1+\frac{1}{\sqrt{3}}+\frac{1}{\sqrt{5}}+....+\frac{1}{\sqrt{119}}\right)$

$\Rightarrow 2(S-1)<2\left(\sqrt{x}\right)_{1}^{121}<2\left(5-\frac{1}{11}\right)$

$\Rightarrow \underbrace{S-1<10}_{S<11}<\underbrace{S-\frac{1}{11}}_{S>10+\frac{1}{11}}\Rightarrow 10+\frac{1}{11}<S<11$

$\Rightarrow$ [S] = 10

I02/1/010. Let a and b be two distinct positive real numbers, then the value of $\int_{a}^{b}\frac{f\left(\frac{x}{a}\right)-f\left(\frac{b}{x}\right)}{x}dx$ (where f(x) is a continuous function) :

माना a तथा b दो भिन्न धनात्मक वास्तविक संख्याएँ हैं, तब $\int_{a}^{b}\frac{f\left(\frac{x}{a}\right)-f\left(\frac{b}{x}\right)}{x}dx$ का मान होगा (जहाँ f(x) एक सतत् फलन है) :

(A) $\frac{1}{a}+\frac{1}{b}$ (B) ab (C) $2\left(\frac{1}{a}-b\right)$ (D) 0

Ans. D

Sol. Let $I=\int_a^b \frac{f\left(\frac{x}{a}\right)-f\left(\frac{b}{x}\right)}{x}dx$

Put $x=\frac{ab}{t}\Rightarrow dx=-\frac{ab}{t^2}dt$

$$I=\int_b^a \frac{\left(f\left(\frac{b}{t}\right)-f\left(\frac{t}{a}\right)\right)}{ab}\left(-\frac{ab}{t^2}\right)dt$$

$$=-\int_a^b \frac{f\left(\frac{t}{a}\right)-f\left(\frac{b}{t}\right)}{t}dt$$

$\therefore$ I = –I $\Rightarrow$ 2I = 0 $\Rightarrow$ I = 0

I02/1/011. Let $F(x)=C\sin x\int_0^x \cos t\,dt+2\int_0^x t.dt+C\cos^2 x-x^2$ (where $C \in R^+$). If $x2 - 2x + 3 \geq F(x)\ \forall\ x \geq R$, then the maximum value of the function $\left(\frac{6e^x-e^{2x}-1}{e^x}\right)^{F(2)}$ is equal to :

माना $F(x)=C\sin x\int_0^x \cos t\,dt+2\int_0^x t.dt+C\cos^2 x-x^2$ (जहाँ $C \in R^+$) है। यदि $x2 - 2x + 3 \geq F(x)\ \forall\ x \geq R$, तब फलन $\left(\frac{6e^x-e^{2x}-1}{e^x}\right)^{F(2)}$ का अधिकतम मान होगा :

(A) 2 (B) 4 (C) 16 (D) 32

Ans. C

Sol. Given $F(x)=C\sin x\int_0^x \cos t\,dt+2\int_0^x t.dt+C\cos^2 x-x^2$

$\Rightarrow F'(x) = 0$ $\therefore$ F(x) is a constant function

$\therefore$ F(x) = F(0) = c

Given $x^2 - 2x + 3 \geq C\ \forall\ x \Rightarrow x^2 - 2x + (3 - 1) \geq 0\ \forall\ x$

$\Rightarrow D \leq 0 \Rightarrow 4 - 4(3 - C) \leq 0$

$\Rightarrow 1 - 3 + C \leq 0 \Rightarrow C \leq 2$

$$\text{Now, } \left(\frac{6e^x - e^{2x} - 1}{e^x}\right)^{f(2)} = \left(\underbrace{6 - \left(e^x + \frac{1}{e^x}\right)}_{\le 4}\right) \le 4^2$$

I02/1/012. A function f(x) satisfy $f(x) = f\left(\frac{4}{x}\right)$ for $x > 0$ if $\int_1^2 \frac{f(x)}{x}dx = 3$, then value of $\int_1^4 \frac{f(x)}{x}dx$ is :

एक फलन f(x), $f(x) = f\left(\frac{4}{x}\right)$ को $x > 0$ के लिए संतुष्ट करता है। यदि $\int_1^2 \frac{f(x)}{x}dx = 3$, तब $\int_1^4 \frac{f(x)}{x}dx$ का मान होगा:

(A) 0 (B) 3 (C) –3 (D) 6

Ans. D

Sol.

$$\int_1^4 \frac{f(x)}{x}dx$$

$$\int_1^2 \frac{f(x)}{x}dx + \int_2^4 \frac{f(x)}{x}dx$$

$$\int_1^2 \frac{f(x)}{x}dx + \int_1^2 \frac{f\left(\frac{4}{x}\right)}{x}dx = 6$$

I03/1/013. Through any point (x, y) of a curve which passes through the origin, lines are drawn parallel to the coordinate axes. The curve, given that it divides the rectangle formed by the two lines and the axis into two areas, one of which is twice the other, represent a family of :

(A) circles (B) parabolas (C) hyperbolas (D) straight lines

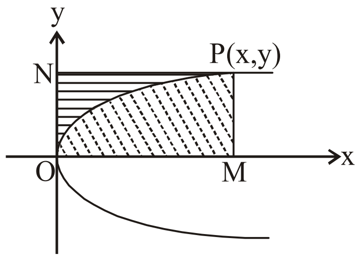
वक्र जो मूल बिन्दु से गुजरता है, के किसी बिन्दु (x, y) से निर्देशी अक्षों के समान्तर रेखाएँ खींची जाती हैं। दिया गया वक्र इन रेखाओं एवं अक्षों से बने आयत को दो क्षेत्रों में विभाजित करता है जिनमें से एक का क्षेत्रफल दूसरे का दुगुना है। तब वक्र निम्न में से किस युग्म को प्रदर्शित करता है :

(A) वृत्त (B) परवलय (C) अतिपरवलय (D) सरल रेखाएँ

Ans. B

Sol. Let P(x, y) be the point on the curve passing through the origin O(0,0), and let PN and PM be the lines parallel to the x and y-axes, respectively. If the equation of the curve is y = y(x), the area of POM equals $\int_0^x y\,dx$ and the area PON equals $xy - \int_0^x y\,dx$. Assuming that 2(POM) = PON, we have

$$2\int_0^x y\,dx = xy - \int_0^x y\,dx \Rightarrow 3\int_0^x y\,dx = xy$$



Differentiating both sides of this gives

$$3y = x\frac{dy}{dx} + y \Rightarrow 2y = x\frac{dy}{dx} \Rightarrow \frac{dy}{y} = 2\frac{dx}{x}$$

$\Rightarrow \log|y| = 2\log x + C \Rightarrow y = Cx^2$, with C being a constant.

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

I04/2/014. Area of the region bounded by the curve $y = \tan x$ and lines $y = 0$ and $x = 1$ is equal to :

वक्र $y = \tan x$ तथा रेखाओं $y = 0$ तथा $x = 1$ के द्वारा परिबद्ध क्षेत्र का क्षेत्रफल होगा :

(A) $\int_0^1 \tan(1-x)\,dx$ (B) $\int_0^{\tan 1} \tan^{-1} y\,dy$ (C) $\int_0^1 \tan^{-1} x\,dx$ (D) $\tan 1 - \int_0^{\tan 1} \tan^{-1} x\,dx$

Ans. AD

Sol. Area $= \int_0^1 \tan x\,dx$

which is also equal to $\int_0^1 \tan(1-x)\,dx$

Area of region OAP = (Area of region OAPB) – (Area of region OPB)

$$= (1\cdot\tan 1) - \int_0^{\tan 1} \tan^{-1} y\,dy$$

$$= \tan 1 - \int_0^{\tan 1} \tan^{-1} x\,dx$$

I04/2/015. A focus of the curve which satisfies the differential equation $(1 + y^2)\,dx - xy\,dy = 0$ and which passes through (1, 0) is $(\pm\sqrt{k}, 0)$. Another curve $y = f(x)$, $(x \geq 0)$ passes through the origin such that $f(1) = k$. Through any point (x, y) on the cuve, lines are drawn parallel to the co-ordinate axes. If the curve (function is concave downward) divides the area formed by these lines and coordinate axes in the ratio 1 : 2 then which of the following is/are correct :

(A) $f(x) > x$ for all $x \in (0,4)$ (B) $f(x) < x$ for all $x \in (0,4)$

(C) $f(x) > x$ for all $x \in (4,\infty)$ (D) $f(x) < x$ for all $x \in (4,\infty)$

अवकलन समीकरण $(1 + y^2)\,dx - xy\,dy = 0$ का हल एक वक्र है, जो (1, 0) से गुजरता है, तथा उसकी नाभि $(\pm\sqrt{k}, 0)$ है। एक वक्र $y = f(x)$, ($x \geq 0$) मूल बिन्दु से इस प्रकार गुजरता है कि $f(1) = k$. वक्र पर स्थित किसी बिन्दु (x, y) से निर्देशांक अक्षों के समान्तर रेखाऐं खींची जाती है। यदि वक्र (फलन नीचे की ओर अवतल है) रेखाओं तथा निदेर्शी अक्षों द्वारा बने क्षेत्रफल को 1 : 2 में विभाजित करता है, तो निम्न में से कौनसा/कौनसे कथन सत्य है :

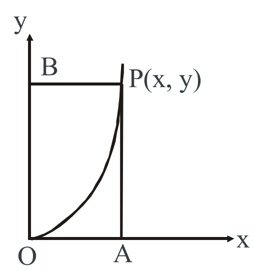
(A) $f(x) > x$ सभी $x \in (0,4)$ (B) $f(x) < x$ सभी $x \in (0,4)$

(C) $f(x) > x$ सभी $x \in (4,\infty)$ (D) $f(x) < x$ सभी $x \in (4,\infty)$

Ans. BC

Sol. Area of OBPO : area of OPAO = m : n

$$\Rightarrow \quad \frac{xy - \int_0^x y\,dx}{\int_0^x y\,dx} = \frac{m}{n}$$



$$\Rightarrow \quad nxy = (m+n)\int_0^x y\,dx$$

Differentiating w.r.t x, we get

$$n\left(x\frac{dy}{dx} + y\right) = (m+n)y$$

$$\Rightarrow \quad nx\frac{dy}{dx} = my \Rightarrow \frac{m}{n}\frac{dx}{x} = \frac{dy}{y}$$

$\Rightarrow \quad y = cx^{m/n}$. (put m =1 and n = 2)

$y = 2\sqrt{x}$

I04/2/016. Area enclosed by parabola $ay = 3(a^2 - x^2)$ and x-axis is 64, a > 0, let y = f(x) be a real valued function satisfying $x\frac{dy}{dx} = x^2 + y - 2$, $f(1) = a - 3$, then :

परवलय $ay = 3(a^2 - x^2)$ तथा x-अक्ष द्वारा परिबद्ध क्षेत्रफल 64 वर्ग इकाई हो, a > 0, माना कि y = f(x) एक वास्तविक फलन

है जो अवकलनीय समीकरण $x\frac{dy}{dx} = x^2 + y - 2$, को संतुष्ट करता है एवं $f(1) = a - 3$, तो :

(A) f(x) is minimum at x = 1 (B) f(x) is maximum at x = 1

(C) f(3) = 5 (D) f(2) = 3

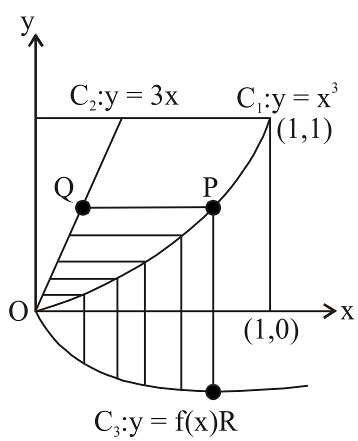
Ans. AC

Sol. $\frac{xdy - ydx}{x^2} = \frac{x^2 - 2}{x^2}dx$

$\int d\left(\frac{y}{x}\right) = \int\left(1 - \frac{2}{x^2}\right)dx$

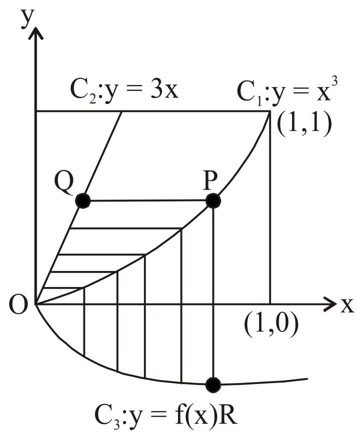
$y = x^2 - 2x + 2$ $(\because f(1) = 1)$

I04/2/017. Let C_1 and C_2 be the graphs of $y = x^3$ and $y = 3x$, $x \in [0, 1]$ respectively. Let C_3 be the graph of a function $y = f(x)$ in [0, 1], f(0) = 0. For a point P on C_1, let the lines through P, parallel to the axes meet C_2 and C_3 at Q and R respectively (see figure). If f(x) is such that for every position of P (on C_1) the areas of shaded regions OPQ & ORP are equal, then $\int_0^1 f(x)dx = -\frac{1}{k}$. If y(x) satisfies the differential equation $\frac{dy}{dx} = \sin 2x + ky\cot x$ and $y\left(\frac{\pi}{2}\right) = 2$ then which of the following statement(s) ia/are correct?



माना $x \in [0, 1]$ के लिए $y = x^3$ तथा $y = 3x$ के वक्र क्रमशः C_1 तथा C_2 है। माना [0, 1] में फलन y = f(x) का ग्राफ C_3 है, f(0) = 0। C_1 पर बिन्दु P से, अक्षों के समान्तर रेखाऐं C_2 तथा C_3 को क्रमशः बिन्दु Q तथा R पर काटती है (चित्र देखे) । यदि f(x) इस प्रकार है कि P (C_1 पर) की प्रत्येक स्थिति के लिए OPQ तथा ORP के छायांकित भाग का क्षेत्रफल बराबर है, तो $\int_0^1 f(x)dx = -\frac{1}{k}$. यदि y(x) अवकल समीकरण $\frac{dy}{dx} = \sin 2x + ky\cot x$ को सन्तुष्ट करता है तथा $y\left(\frac{\pi}{2}\right) = 2$ है तो निम्न

में से कौनसा/कौनसे कथन सही है :



(A) $y\left(\frac{\pi}{6}\right) = 0$

(B) $y'\left(\frac{\pi}{3}\right) = \frac{9-3\sqrt{2}}{2}$

(C) y(x) अंतराल $\left(\frac{\pi}{6},\frac{\pi}{3}\right)$ में वर्धमान है

(D) $\int_{-\pi/2}^{\pi/2} y(x)dx = \pi$

Ans. AC

Sol. Let P (α, α^3)

$$\int_0^{\alpha}\left(x^3 - f(x)\right)dx = \int_0^{\alpha^3}\left(y^{1/3} - \frac{y}{3}\right)dy$$

$$\alpha^3 - f(\alpha) = \left(\alpha - \frac{\alpha^3}{3}\right)3\alpha^2$$

$f(a) = \alpha^5 - 2\alpha - 2\alpha^3 \Rightarrow f(x) = x^5 - 2x^3$

(A) $\int_0^1\left(x^2 - 2x^3\right)dx = \frac{1}{6} - \frac{1}{2} = \frac{1}{3}$

(B) $f\left(\frac{1}{2}\right) = \frac{1}{32} - \frac{1}{4} = -\frac{7}{32}$

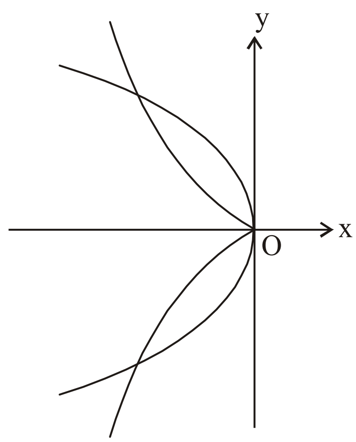
(D) $f'(x) = x^2(5x^2 - 6) < 0 \ \forall x \in (0,1)$

I04/2/018. Area of region enclosed between curves given by $y^2 = -4x$ and $-x = \sqrt{|y|}$ is 3k (in sq. units). If α, β are the roots of $4x^2 - 2kx + c = 0$, $c > 0$ such that $1 < \alpha < 2 < \beta < 3$, then value of 'c' can be :

वक्रो $y^2 = -4x$ तथा $-x = \sqrt{|y|}$ के मध्य क्षेत्र का क्षेत्रफल 3k होगा (वर्ग इकाई में) यदि α, β समीकरण $4x^2 - 2kx + c = 0$, $c > 0$ के मूल है इस प्रकार की $1 < \alpha < 2 < \beta < 3$, तो 'c' का मान हो सकता है :

(A) 11 (B) 12 (C) 13 (D) 14

Ans. CD



Sol. $A = 2 \times \frac{16}{3}(1)\left(\frac{1}{4}\right) = \frac{8}{3}$

I04/2/019. Area enclosed by the curve $y = (x^2 + 2x)e^{-x}$ and the positive x-axis is m. The value of k for which both roots of the equation $mx^2 - 2x + k = 0$ are completely in $(-1, 1)$, may be equal to :

धनात्मक x-अक्ष तथा वक्र $y = (x^2 + 2x)e^{-x}$ द्वारा घिरा क्षेत्रफल m होगा। k का मान ज्ञात कीजिए जिसके लिए समीकरण $mx^2 - 2x + k = 0$ के दोनों मूल पूर्णतया $(-1, 1)$, के मध्य होंगे :

(A) –1 (B) 0 (C) 2 (D) –3

Ans. AB

Sol. $m = \int_0^{\infty}\left(x^2 + 2x\right)e^{-x}dx$

I04/2/020. The area (in sq. units) of the region consisting of points (x, y) on xy plane which satisfy $|x| \le 1 + |y|$ and $|y| \le 1$ is 2k. Let T_r be the r^{th} term of a sequence, for r = 1, 2, 3, 4, ... If $(k\,T_{r+1} = T_r)$ and $T_7 = \frac{1}{243}$, then :

xy समतल में $|x| \le 1 + |y|$ तथा $|y| \le 1$ द्वारा परिबद्ध क्षेत्रफल 2k है। माना r = 1, 2, 3, 4, ... के लिए T_r श्रेणी का r वां पद

है। यदि $(k\,T_{r+1} = T_r)$ तथा $T_7 = \dfrac{1}{243}$, तो :

(A) $\sqrt{\sum_{r=1}^{5}\dfrac{1}{T_{r+1}}} = 9$ (B) $\sqrt{\sum_{r=1}^{5}\dfrac{1}{T_{r+1}}} = 11$ (C) $\sum_{r=1}^{\infty}\left(T_r.T_{r+1}\right) = \dfrac{27}{8}$ (D) $\sum_{r=1}^{\infty}\left(T_r.T_{r+1}\right) = \dfrac{9}{2}$

Ans. BC

Sol. $|y| \le 1 \Rightarrow y \in [-1, 1]$

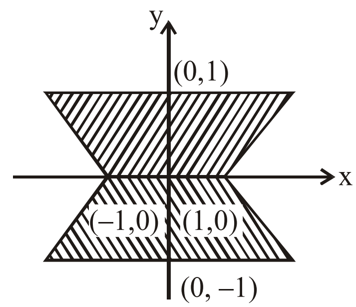
$|x| - |y| \le 1 \qquad \Rightarrow x - y \le 1\ (x > 0, y > 0)$

$\Rightarrow x + y \le 1\ (x > 0, y < 0)$

$\Rightarrow -x - y \le 1\ (x < 0, y > 0)$

$\Rightarrow -x + y \le 1\ (x < 0, y < 0)$

Area of shaded region is 6 sq. units.



$$\frac{T_{r+1}}{T_r} = \frac{1}{3}$$

$T_1 = a,\ T_2 = \dfrac{a}{3},\ T = \dfrac{a}{3^2}$ and so on

$$\therefore\ T_7 = \frac{a}{3^6} = \frac{1}{243} \Rightarrow a = 3$$

Now, $T_r.T_{r+1} = \left(\dfrac{a}{3^{r-1}}\right)\left(\dfrac{a}{3^r}\right) = \dfrac{3^2}{3^{2r+1}}$

Also, $\sum_{r=1}^{5}\dfrac{1}{T_{r+1}} = \sum_{r=1}^{5}\dfrac{3^r}{a} = \sum_{r=1}^{5}3^{r-1}$

$= (1 + 3 + 3^2 + 3^3 + 3^4) = 121$

$$\Rightarrow \sqrt{\sum_{r=1}^{5}\frac{1}{T_{r+1}}} = 11.$$

and $\sum_{r=1}^{\infty} T_r \cdot T_{r+1} = \dfrac{\left(3 + \dfrac{1}{3} + \dfrac{1}{3^3} + \dfrac{1}{3^5} + ...\infty\right)}{\text{infinite G.P.}}$

$$= \frac{3}{1 - \dfrac{1}{3^2}} = \frac{3}{\dfrac{8}{9}} = \frac{27}{8}$$

I04/2/021. If A is the area bounded by curves $y = |\cos x|$, $y = 5 - \dfrac{4}{\pi}|x - \pi|$, in $x \in [0, 2\pi]$, then value of $\left(\dfrac{A}{2} + 2\right)$ is (k

$-1)\pi$. The sum $\sum_{n=1}^{\infty}\left(\frac{n}{n^4+k}\right)$ is equal to :

यदि A अन्तराल $x \in [0, 2\pi]$ में वक्रों $y = |\cos x|$, $y = 5-\frac{4}{\pi}|x-\pi|$ द्वारा परिबद्ध क्षेत्रफल हो, तो $\left(\frac{A}{2}+2\right)$ का मान (k

$-1)\pi$ होगा। Øè» $\sum_{n=1}^{\infty}\left(\frac{n}{n^4+k}\right)$ ·¤æ ×æÙ ãæð»æ :

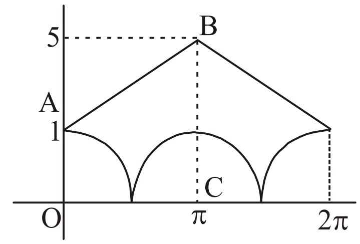
(A) 1/4 (B) 1/3 (C) 3/8 (D) $\frac{1}{2.\overline{6}}$

Ans. CD

Sol. Required Area = 2(Area of Trapezium OABC – 2)

$$= 2\left\{\frac{1}{2}.6.\pi - 2\right\} = 6\pi - 4 = A$$

$$\sum_{n=1}^{\infty}\left(\frac{n}{n^4+4}\right) = \sum_{n=1}^{\infty}\frac{1}{4}\left(\frac{1}{n^2-2n+2} - \frac{1}{n^2+2n+2}\right)$$



I04/2/022. Let $f(x) = \text{Min.}(x^2+1,\ x+1)$ for $x \in [0,2]$. If A is the area bounded by $f(x)$, x-axis, in the interval [0, 2], then the value of 6A is 10k + m. The volume of the parallelopiped whose coterminous edges are represented by the vectors $k\vec{b}\times\vec{c}, m\,\vec{c}\times\vec{a}$ and $4\vec{a}\times\vec{b}$ where $\vec{a} = (1+\sin\theta)\hat{j} + \sin 2\theta\hat{k}$,

$$\vec{b} = \sin\left(\theta+\frac{2\pi}{3}\right)\hat{i} + \cos\left(\theta+\frac{2\pi}{3}\right)\hat{j} + \sin\left(2\theta+\frac{4\pi}{3}\right)\hat{k},$$

$\vec{c} = \sin\left(\theta-\frac{2\pi}{3}\right)\hat{i} + \cos\left(\theta-\frac{2\pi}{3}\right)\hat{j} + \sin\left(2\theta-\frac{4\pi}{3}\right)\hat{k}$ is 18 cubic units, then the values of θ, in the interval $\left(0, \frac{\pi}{2}\right)$; is/are :

माना $f(x) = \text{Min.}(x^2+1,\ x+1)$, जहाँ $x \in [0, 2]$ है। यदि A, अन्तराल [0, 2] में $f(x)$, x-अक्ष द्वारा परिबद्ध क्षेत्रफल है, तो 6A का मान 10k + m है।समान्तर षट्फलक का आयतन, जिसके संगामी कोर (edges) $k\vec{b}\times\vec{c}, m\,\vec{c}\times\vec{a}$ तथा $4\vec{a}\times\vec{b}$ द्वारा दिये जाते हैं जहाँ $\vec{a} = (1+\sin\theta)\hat{j} + \sin 2\theta\hat{k}$, $\vec{b} = \sin\left(\theta+\frac{2\pi}{3}\right)\hat{i} + \cos\left(\theta+\frac{2\pi}{3}\right)\hat{j} + \sin\left(2\theta+\frac{4\pi}{3}\right)\hat{k}$,

$\vec{c} = \sin\left(\theta - \frac{2\pi}{3}\right)\hat{i} + \cos\left(\theta - \frac{2\pi}{3}\right)\hat{j} + \sin\left(2\theta - \frac{4\pi}{3}\right)\hat{k}$, 18 घन इकाई है तो θ का मान $\left(0, \frac{\pi}{2}\right)$ अन्तराल में होगा :

(A) $\frac{\pi}{9}$ (B) $\frac{2\pi}{9}$ (C) $\frac{\pi}{3}$ (D) $\frac{4\pi}{9}$

Ans. ABD

Sol. Value $= \left|\left[2\vec{b}\times\vec{c}, 3\vec{c}\times\vec{a}, 4\vec{a}\times\vec{b}\right]\right| = 18$

$24\left[\vec{a}\vec{b}\vec{c}\right] = 18 \Rightarrow \left|\left[\vec{a}\vec{b}\vec{c}\right]\right| = \frac{\sqrt{3}}{2}$

$$\text{Now, } \left[\vec{a}\vec{b}\vec{c}\right] = \begin{vmatrix} (1+\sin\theta) & \cos\theta & \sin 2\theta \\ \sin\left(\theta + \frac{2\pi}{3}\right) & \cos\left(\theta + \frac{2\pi}{3}\right) & \sin\left(2\theta + \frac{4\pi}{3}\right) \\ \sin\left(\theta - \frac{2\pi}{3}\right) & \cos\left(\theta - \frac{2\pi}{3}\right) & \sin\left(2\theta - \frac{4\pi}{3}\right) \end{vmatrix}$$

Applying $R_1 + R_1 + R_2 + R_3$ we get after simplification,

$$\left|\left[\vec{a}\vec{b}\vec{c}\right]\right| = \sqrt{3}\,|\cos 3\theta| = \frac{\sqrt{3}}{2} \Rightarrow \cos 3\theta = \pm\frac{1}{2} \Rightarrow 3\theta = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3} \Rightarrow \theta = \frac{\pi}{9}, \frac{2\pi}{9}, \frac{4\pi}{9}$$

I04/2/023. The area bounded by the curve $y^2 + x^4 = x^2$ is 3k. Given that $\vec{a}$ and $\vec{b}$ are two unit vectors such that angle between $\vec{a}$ and $\vec{b}$ is $\cos^{-1}\left(\frac{1}{4}\right)$. If $\vec{c}$ be a vector in the plane of $\vec{a}$ and $\vec{b}$, such that $|\vec{c}| = k$, $\vec{c}\times\vec{b} = 2\vec{a}\times\vec{b}$ and $\vec{c} = \lambda\vec{a} + \mu\vec{b}$ then :

(A) $\lambda = 2$ (B) sum of values of $\mu = -1$

(C) product of all possible values of $\mu = -12$ (D) number of distinct values of μ is three.

वक्र $y^2 + x^4 = x^2$ से घिरा हुआ क्षेत्रफल 3k होगा। दिया गया है कि $\vec{a}$ तथा $\vec{b}$ दो इकाई सदिश है इस प्रकार कि $\vec{a}$ तथा $\vec{b}$ के मध्य कोण $\cos^{-1}\left(\frac{1}{4}\right)$ है। यदि $\vec{c}$ सदिश $\vec{a}$ तथा $\vec{b}$ के तल में एक सदिश है इस प्रकार कि $|\vec{c}| = k$, $\vec{c}\times\vec{b} = 2\vec{a}\times\vec{b}$ तथा $\vec{c} = \lambda\vec{a} + \mu\vec{b}$ है तो :

(A) $\lambda = 2$ (B) μ के मानों का योग – 1 है

(C) μ के सभी सम्भव मानों का गुणन – 12 है (D) μ के भिन्न मानों की संख्या 3 है

Ans. ABC

Sol. Given $\vec{c}\times\vec{b} = 2\vec{a}\times\vec{b}$

$\Rightarrow \quad (2\vec{a}-\vec{c})\times\vec{b}=\vec{0} \Rightarrow \quad (2\vec{a}-\vec{c})\parallel\vec{b} \quad \Rightarrow \quad \vec{c}=2\vec{a}+\mu\vec{b} \quad ....(1)$

comparing with $\vec{c}=\lambda\vec{a}+\mu\vec{b}$

we have $\boxed{\lambda=2}$ hence (A) is correct.

$\Rightarrow$ from equation (1) $\vec{a}\cdot\vec{c} = 2\vec{a}\cdot\vec{a}+\mu\vec{a}\cdot\vec{b}$ [take dot with $\vec{a}$ in (1)]

$\Rightarrow \quad \vec{a}\cdot\vec{c} = 2+\mu\,\frac{1}{4} \quad ....(2)$

{using $|\vec{a}|=|\vec{b}|=1$ and angle between $\vec{a}\,\&\,\vec{b} = \cos^{-1}\frac{1}{4}$ }

from equation (1), again $\vec{b}\cdot\vec{c} = 2\vec{a}\cdot\vec{b}+\mu\vec{b}\cdot\vec{b}$ [take dot with $\vec{b}$ in (1)]

$\Rightarrow \quad \vec{b}\cdot\vec{c} = 2\,\frac{1}{4}+\mu\,1 \quad ....(3)$

{using $|\vec{a}| = |\vec{b}| = 1$ and angle between $\vec{a}\,\&\,\vec{b} = \cos^{-1}\left(\frac{1}{4}\right)$}

again from equation (1) dot with $\vec{c}$

$\vec{c}\cdot\vec{c} = 2\vec{a}\cdot\vec{c}+\mu\vec{b}\cdot\vec{c}$

from equation (2) and (3) (using $|\vec{c}|=4$)

$$4^2 = 2\left(2+\mu\,\frac{1}{4}\right)+\mu\left(\frac{2}{4}+\mu\right)$$

$$16 = 4+\frac{\mu}{2}+\frac{\mu}{2}+\mu^2 \Rightarrow \mu^2+\mu-12=0 \left\langle \begin{matrix}\mu=3\\ \mu=-4\end{matrix}\right.$$

$\therefore$ sum of values of $\mu = -1$ & product of values of $\mu = -12$ $\quad\therefore$ (B) and (C) are also correct

I04/2/024. If area bounded by $y=\dfrac{a}{4+x^2}$ $(a>0)$ and $y=c(c\le 0)$ is an even integer :

यदि $y=\dfrac{a}{4+x^2}$ $(a>0)$ तथा $y=c(c\le 0)$ से परिबद्ध क्षेत्रफल एक सम अभाज्य पूर्णांक है :

(A) $a=\dfrac{4}{\pi}$ (B) $a=2$ (C) $c=1$ (D) $c=0$

Ans. AD

Sol. $y=\frac{a}{4+x^2}$

$\text{Area} = 2\int_0^{\infty}\frac{a}{4+x^2}dx$

$=2a\left[\frac{1}{2}\tan^{-1}\frac{x}{2}\right]_0^{\infty}$

$\Rightarrow \frac{\pi a}{2}$

I04/2/025. Consider $f(x)=\begin{cases} x^{\frac{1}{3}} & ; \quad 0\le x<1 \\ (x-2)^2 & ; \quad 1\le x<2 \end{cases}$, and f(–x) = f(x) $\forall$ x $\in$ R and f(x + 2) = f(x) $\forall$ x $\ge$ 0, then identify the correct statement(s) :

(A) The number of points of non-derivability of f(x) in [0, 10] is(are) 11

(B) The number of points of non-derivability of f(x) in [0, 10] is(are) 9

(C) Area of the region enclosed by the f(x) and the x-axis from x = –1 to x = 7 is $\frac{19}{4}$ sq. units

(D) Area of the region enclosed by the f(x) and the x-axis from x = –1 to x = 7 is $\frac{49}{12}$ sq. units

माना कि $f(x)=\begin{cases} x^{\frac{1}{3}} & ; \quad 0\le x<1 \\ (x-2)^2 & ; \quad 1\le x<2 \end{cases}$, तथा f(–x) = f(x) $\forall$ x $\in$ R और f(x + 2) = f(x) $\forall$ x $\ge$ 0, सही कथन/कथनों को पहचानिए :

(A) [0, 10] में f(x) के अवकलनीय नहीं होने वाले बिन्दुओं की संख्या 11 है

(B) [0, 10] में f(x) के अवकलनीय नहीं होने वाले बिन्दुओं की संख्या 11 है

(C) वक्र f(x) और x-अक्ष का x = –1 से x = 7 से परिबद्ध क्षेत्रफल $\frac{19}{4}$ वर्ग इकाई है

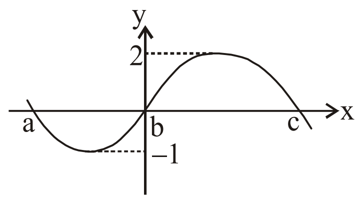
(D) वक्र f(x) और x-अक्ष का x = –1 से x = 7 से परिबद्ध क्षेत्रफल $\frac{49}{12}$ वर्ग इकाई है

Ans. AC

Sol.

I04/2/026. Let f(x) be a polynomial function of degree 3 where a < b < c and f(a) = f(b) = f(c). If the graph of f(x) is as shown, which of the following statements are **INCORRECT** ? (Where $c > |a|$)

माना f(x) 3 कोटि का एक बहुपद है जहाँ $a < b < c$ तथा $f(a) = f(b) = f(c)$ हैं। यदि f(x) का रेखाचित्र दर्शाया गया है, तो निम्न कथनों में से कौनसा/से कथन असत्य होगा/होंगे ? (जहाँ $c > |a|$)



(A) $\int_a^c f(x)dx = \int_b^c f(x)dx + \int_a^b f(x)dx$ (B) $\int_a^c f(x)dx < 0$

(C) $\int_a^b f(x)dx < \int_c^b f(x)dx$ (D) $\frac{1}{b-a}\int_a^b f(x)dx > \frac{1}{c-b}\int_b^c f(x)dx$

Ans. bcd

Sol. $f(x) = -(x-a)(x-b)(x+c) = (x-a)(x)(x+c)$ $[\because\ b = 0]$

clearly option (A) is correct

(B) $\int_a^c f(x)dx = \int_a^b f(x)dx + \int_b^c f(x)dx > 0$ (from grest) which incorrect

(C) $\int_a^b f(x)dx < 0$ & $\int_c^b f(x)dx < 0$ but second term is large negative value so option (c) is incorrect

(D) clearly D is incorrect.

I02/2/027. Let f be a differentiable function satisfying the integral equation $\int_0^x f(t)dt + \int_0^x t.f(x-t)dt = e^{-x} - 1$ then which of the following is/are true :

माना f अवकलनीय फलन, समाकल समीकरण $\int_0^x f(t)dt + \int_0^x t.f(x-t)dt = e^{-x} - 1$ को संतुष्ट करता है, तो निम्न में से कौनसा/कौनसे विकल्प सत्य होगा/होगें :

(A) $f(2) = \frac{1}{e^2}$ (B) $f(0) + f'(0) = 1$ (C) $f'(0) = 2$ (D) $f'(0) = 1$

Ans. ABC

Sol. $\int_0^x f(t)dt + \int_0^x t\cdot f(x-t)dt = e^{-x} - 1$

$$\Rightarrow \int_0^x f(t)dt + x\int_0^x f(t)dt - \int_0^x t.f(t)dt = e^{-x} - 1$$

diff. with respect to x.

$$f(x) + x \cdot f(x) + \int_0^x f(t)dt - x \cdot f(x) = -e^{-x}$$

$$\Rightarrow \left(e^x \mid \int_0^{-} f(t)dt + f(x)\right) = -1 \text{ on integrating}$$

$$e^x \int_0^x f(t)dt = -x + c \quad \text{put } x = 0 \Rightarrow c = 0$$

$$\therefore e^x \int_0^x f(t)dt = -x \text{ again diff. w.r. to x.}$$

$$e^x \int_0^x f(x) + e^x \int_0^x f(t)dt = -1$$

$$\text{using (1)}, e^x \cdot f(x) - x = -1 \Rightarrow f(x) = \frac{x-1}{e^x}$$

I02/2/028. Let $f : R \to R$ is defined by $f(x) = x + \sin x$ & $\int_0^{\pi} f^{-1}(x)dx = I$, then :

माना $f : R \to R$ फलन $f(x) = x + \sin x$ से परिभाषित है और $\int_0^{\pi} f^{-1}(x)dx = I$ है, तो :

(A) $I > \int_0^1 \frac{dx}{1+x^3}$ (B) $I < \int_0^1 e^{x^2}dx$ (C) $2 < I < 3$ (D) $\frac{\pi}{4} < I < \frac{\pi}{2}$

Ans. AC

Sol. $I = \int_0^{\pi} f^{-1}(x)dx$

Let $x = f(x) \quad \Rightarrow \quad dx = f'(t)dt$

$$\Rightarrow I = \int_0^{\pi} tf'(t)dt = \frac{\pi^2}{2} - 2 \approx 2.89$$

$$\text{Obviously } \int_0^1 \frac{dx}{1+x^3} < \int_0^1 dx$$

$$\Rightarrow \int_0^1 \frac{dx}{1+x^3} < 1 \Rightarrow \text{(A) is true}$$

I02/2/029. Let $I = \int\limits_{-\pi/3}^{\pi/3} (\cos x)^{\sin x} dx$, then I is greater than :

माना $I = \int\limits_{-\pi/3}^{\pi/3} (\cos x)^{\sin x} dx$ हो, तो I निम्न से अधिक होगा :

(A) $\frac{\pi}{2}$ (B) $\frac{2\pi}{3}$ (C) $\frac{4\pi}{3}$ (D) 2π

Ans. AB

Sol. $I = \int_{-\pi/3}^{\pi/3} (\cos x)^{\sin x} dx$ (apply king)

$I = \int_{-\pi/3}^{\pi/3} (\cos x)^{-\sin x} dx$ (add)

$$2I = \int_{-\pi/3}^{\pi/3} \left((\cos x)^{\sin x} + \frac{1}{(\cos x)^{\sin x}} \right) dx$$

$$(\cos x)^{\sin x} + \frac{1}{(\cos x)^{\sin x}} \geq 2$$

$$2I > \int_{-\pi/3}^{\pi/3} 2dx \Rightarrow 2I > \frac{4\pi}{3} \Rightarrow I > \frac{2\pi}{3}$$

but always less than $\frac{4\pi}{3}$

$\because \cos x^{\sin x} \leq 2^{(\sqrt{3}/2)}$

I02/2/030. If $I = \int\limits_{0}^{1/2} \frac{dx}{\sqrt{1-x^{2n}}}, n \in N$, then :

यदि $I = \int\limits_{0}^{1/2} \frac{dx}{\sqrt{1-x^{2n}}}, n \in N$ हो, तब :

(A) $I \geq \frac{1}{2}$ (B) $I \leq \frac{\pi}{6}$ (C) $I \geq \frac{\pi}{4}$ (D) $I \leq \frac{\pi}{2}$

Ans. ABD

Sol. $1 \leq \sqrt{1-x^{2n}} \leq \sqrt{1-x^2}$

$$\frac{1}{\sqrt{1-x^2}} \leq \frac{1}{\sqrt{1-x^{2n}}} \leq 1$$

$$\int_0^{1/2} \frac{1}{\sqrt{1-x^2}} dx \le \int_0^{1/2} \frac{1}{\sqrt{1-x^{2n}}} \le \int_0^{1/2} 1 dx$$

$$\frac{\pi}{6} \le I \le \frac{1}{2}$$

I02/2/031. Let $f : R \to R$ be a real valued differentiable function satisfying the integral equation $1+\int_1^x 2t^2 f(t)dt = x^2 + 2\int_x^1 f(t)dt$, then which of the following is (are) true :

(A) $f(x)$ has exactly one asymptote

(B) $f(x)$ has neither injective nor surjective

(C) Distance between two horizontal tangents of $f(x)$ is 1

(D) If the equation $f(|x|) = k$ has four distinct solutions, then range of k is equal to $\left(0, \frac{1}{2}\right]$

माना $f : R \to R$ वास्तविक मान अवकलनीय फलन, समाकल समीकरण $1+\int_1^x 2t^2 f(t)dt = x^2 + 2\int_x^1 f(t)dt$ को संतुष्ट करता हैं, तो निम्न में से कौनसा/कौनसे कथन सत्य होगा/होगें ?

(A) $f(x)$ की केवल एक अनन्तस्पर्शी है

(B) $f(x)$ ना तो एकैकी ना ही आच्छादक है

(C) $f(x)$ की दो क्षैतिज स्पर्श रेखाओं के मध्य दूरी 1 इकाई है

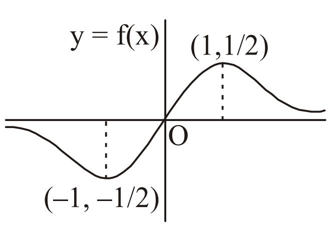
(D) यदि समीकरण $f(|x|) = k$ के चार विभिन्न हल हो, तो k का परिसर $\left(0, \frac{1}{2}\right]$ होगा

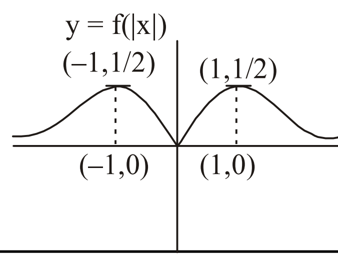
Ans. ABC

Sol. Given equation $1+\int_1^x 2t^2 f(t)dt = x^2 + 2\int_x^1 f(t)dt$

diff. w.r.t. x

$$2x^2 f(x) = 2x + (0 - 2f(x)) \Rightarrow f(x) = \frac{x}{1+x^2}$$





clearly $y=\frac{1}{2}$ is only the horizontal tangent of f(|x|)

and f(|x|) = k has four solutions for $k \in \left(0, \frac{1}{2}\right)$

I02/2/032. Given $f(x)=\int_0^x (6t^4-\alpha t^3)\,dt$ and g(x) is a quadratic polynomial such that g(0) = –9, g'(0) = γ, g"(0) = –2β. Let h(x) = f '(x) and the graph of y = h(x) and y = g(x) intersect in four distinct points with positive abscissae x_i ; i = 1, 2, 3, 4 such that $\sum \frac{i}{x_i}=8$, then $\sum x_i$ is greater than or equal to :

दिया है $f(x)=\int_0^x (6t^4-\alpha t^3)\,dt$ तथा g(x) एक द्विघात बहुपद इस प्रकार है कि g(0) = –9, g'(0) = γ, g"(0) = –2β है। माना h(x) = f '(x) तथा y = h(x) एवं y = g(x) के आरेख चार विभिन्न बिन्दुओं पर प्रतिच्छेद करतें है, जिनके भुज x_i ; i = 1, 2, 3, 4 धनात्मक है तथा $\sum \frac{i}{x_i}=8$ हो, तो $\sum x_i$ का मान निम्न से बड़ा या बराबर होगा :

(A) 2 (B) 4 (C) 5 (D) 7

Ans. ABC

Sol. $\because g(x)=\frac{x^2}{\lfloor 2}g''(0)+\frac{x}{\lfloor 1}g'(0)+g(0)=-\beta x^2+\gamma x-9$

also, $h(x)=f'(x)=6\times 4-\alpha x^3$

Now $h(x)=g(x)\Rightarrow 6x^4-\alpha x^3+\beta x^2-\gamma x+9=0$

By Descartes rule of sign, above biquadratic has four distinct positive roots x_1, x_2, x_3, x_4.

Using A.M. ≥ G.M.

we have $\dfrac{\frac{1}{x_1}+\frac{2}{x_2}+\frac{3}{x_3}+\frac{4}{x_4}}{4}\ge \sqrt[4]{\dfrac{24}{x_1x_2x_3x_4}}$

$\Rightarrow 2\ge 2 \therefore \frac{1}{x_1}=\frac{2}{x_2}=\frac{3}{x_3}=\frac{4}{x_4}=k.$

$\Rightarrow \frac{24}{x_1x_2x_3x_4}=k^4\Rightarrow k^4=\frac{24}{3/2}=16\Rightarrow k=2$

$\therefore x_1=\frac{1}{2}, x_2=1, x_3=\frac{3}{2}, x_4=2$

$\therefore \Sigma x_i = 5$

I02/2/033. If $f(x) = \sin x + \int_{-\pi/2}^{\pi/2} (\sin x + t\cos x) f(t)dt$ can be expressed as $-\frac{1}{k}\sin x - \frac{2}{k}\cos x$, where k is a constant, then k can not be more than :

यदि $f(x) = \sin x + \int_{-\pi/2}^{\pi/2} (\sin x + t\cos x) f(t)dt$ को $-\frac{1}{k}\sin x - \frac{2}{k}\cos x$ के रूप मे व्यक्त किया जा सकता हो, जहाँ k एक नियतांक है, तो k का मान किस से अधिक नहीं हो सकता :

(A) 2 (B) 3 (C) 4 (D) 5

Ans. BCD

Sol. $f(x) = \sin x + \int_{-\pi/2}^{\pi/2} (\sin x + t\cos x)\left(-\frac{1}{k}\sin t - \frac{2}{k}\cos t\right)dt$

$f(x) = \sin x + I_1 + I_2 + I_3 + I_4$

where $I_1 = -\frac{\sin x}{k}\int_{-\pi/2}^{\pi/2} \sin t dt = 0$

$I_2 = -\frac{2\sin x}{k}\int_{-\pi/2}^{\pi/2} \cos t dt = -\frac{4\sin x}{k}$

Similarly $I_3 = -\frac{2\cos k}{k}$ & $I_4 = 0$

$\Rightarrow \quad -\frac{1}{k}\sin x - \frac{2}{k}\cos x = \sin x - \frac{4\sin x}{k} - \frac{2\cos x}{k}$

$\Rightarrow k = 3$

I02/2/034. If the probability that randomly selected point inside area bounded by parabola $y^2 = 8x$ and its latus rectum lies inside the ellipse $\frac{x^2}{2} + \frac{y^2}{16} = 1$ is $\frac{\sqrt{2}(a\pi + b)}{32}$ then. The points of extremum of $\int_0^{x^2}\left(\frac{t^2 - (a+b)t + 4}{2 + e^t}\right)dt$ are :

यदि परवलय $y^2 = 8x$ तथा इसके नाभिलम्ब द्वारा परिबद्ध क्षेत्रफल के अन्दर यादृच्छया चयन किये गये बिन्दु के दीर्घवृत $\frac{x^2}{2} + \frac{y^2}{16} = 1$ के अन्दर स्थित होने की प्रायिकता $\frac{\sqrt{2}(a\pi + b)}{32}$ हो, तो $\int_0^{x^2}\left(\frac{t^2 - (a+b)t + 4}{2 + e^t}\right)dt$ के चरम बिन्दु होगें :

(A) $x = -2$ (B) $x = 1$ (C) $x = 0$ (D) $x = -1$

Ans. ABCD

Sol. Let $f(x)=\int_0^{x^2}\left(\frac{t^2-5t+4}{2+e^t}\right)dt \quad \therefore\ f'(x)=\left(\frac{x^4-5x^2+4}{2+e^{x^2}}\right)\times 2x$

For extremum $f'(x)=0$

I02/2/035. Which of the following statement(s) is(are) correct ?

(A) $\lim_{n\to\infty}\left(\frac{1}{\sqrt{4n-1^2}}+\frac{1}{\sqrt{8n-2^2}}+\frac{1}{\sqrt{12n-3^2}}+\ldots\ldots\ldots+\frac{1}{2n}\right)$ is equal to $\frac{\pi}{2}$

(B) If for a twice differentiable function f(x), $f''(x)<0\ \forall\ x\in(a,b)$ then $f'(x)=0$ has atmost one solution in (a, b)

(C) If $f(x)=ax^7+bx^5+cx^3+dx+2$, where a, b, c are real constants and $f(-3)=3$ then range of $f(3)+3\cos^2x+4\sin^2x$, is equal to [4, 5]

(D) $\ell n(1+x)<\frac{\tan^{-1}x}{1+x}$ for $x>0$

निम्नलिखित में से कौनसा/से विकल्प सत्य होगा/होंगे ?

(A) $\lim_{n\to\infty}\left(\frac{1}{\sqrt{4n-1^2}}+\frac{1}{\sqrt{8n-2^2}}+\frac{1}{\sqrt{12n-3^2}}+\ldots\ldots\ldots+\frac{1}{2n}\right)$ का मान $\frac{\pi}{2}$ है

(B) यदि किसी द्विअवकलनीय फलन f(x) के लिए $f''(x)<0\ \forall\ x\in(a,b)$ है तब $f'(x)=0$ का अंतराल (a, b) में अधिकतम एक हल होगा

(C) यदि $f(x)=ax^7+bx^5+cx^3+dx+2$, जहाँ a, b, c वास्तविक अचर हैं तथा $f(-3)=3$ तब $f(3)+3\cos^2x+4\sin^2x$ का परिसर [4, 5] होगा

(D) $\ell n(1+x)<\frac{\tan^{-1}x}{1+x}$ for $x>0$

Ans. ABC

Sol. (A) $\lim_{n\to\infty}\sum_{r=1}^{2n}\frac{1}{\sqrt{4rn-r^2}}=\int_0^2\frac{dx}{\sqrt{4x-x^2}}=\frac{\pi}{2}$

(B) If $f''(x)<0\ \forall\ x\in(a,b)$ then $f'(x)$ is decreasing in $x\in(a,b)$

$\Rightarrow f'(x)=0$ has atmost one solution.

(C) $f(-3)=-a(3)^7-b(3)^5-c(3)^3-3d+2=3$

$\Rightarrow\ f(3)=a(3)^7+b(3)^5+c(3)^3+3d+2=1$

$1+3\cos^2x+4\sin^2x=5-\cos^2x\in[4,5]$

(D) Let $f(x)=(1+x)\,\ell n(1+x)-\tan^{-1}x$

$f(0)=0$

$f'(x)=\ell n(1+x)+\frac{x^2}{1+x^2}>0\ \forall\ x>0$

$\Rightarrow\ f(x)=(1+x)\,\ell n(1+x)-\tan^{-1}x>0$

I02/2/036. Let $I_n = \int_0^n 2^x \left(1+\{g(x)\}+\{\sqrt{g(x)}\}\right)dx$, $n \in N$, where $g(x) = \sin^2(\pi x)$ and $\{x\}$ represents fractional part of x, then :

माना $I_n = \int_0^n 2^x \left(1+\{g(x)\}+\{\sqrt{g(x)}\}\right)dx$, $n \in N$, जहाँ $g(x) = \sin^2(\pi x)$ तथा $\{x\}$ भिन्नात्मक भाग x फलन है, तब :

(A) $I_1 > \log_2 e$ (B) $I_1 < \log_2 e$ (C) $I_4 = 15I_1$ (D) $I_4 = 16I_1$

Ans. AC

Sol. $I_1 = \int_0^1 2^x \left(1+\{\sin 2\pi x\}+\{|\sin x|\}\right)dx \ge \int_0^1 2^x dx = \left.\frac{2^x}{\ell n 2}\right|_0^1 = \log_2 e$

Let $f(x) = 2x\left(1+\{g(x)\}+\{\sqrt{g(x)}\}\right)$

So, $I_4 = \int_0^4 f(x)dx = \int_0^1 \left(f(x)+f(x+1)+f(x+2)+f(x+3)\right)dx$

$= (1 + 2 + 2^2 + 2^3)\int_0^1 f(x)dx$ (as $g(x + 1) = g(x)$)

$= 15I_1$

I02/2/037. If f(x) & g(x) are two twice differentiable functions satisfying $f''(x) = g''(x)$, $f'(1) = 2g'(1) = 4$ & $f(2) = 3g(2) = 9$, then :

(A) $f(4) - g(4) = 10$ (B) $|f(x) - g(x)| < 2 \Rightarrow -2 < x < 0$

(C) $f(x) = g(x) \Rightarrow x = -1$ (D) $f(x) - g(x) = 2x$ has no real roots

यदि दो फलन f(x) तथा g(x), दो बार अवकलनीय फलन इस प्रकार है कि $f''(x) = g''(x)$, $f'(1) = 2g'(1) = 4$ तथा $f(2) = 3g(2) = 9$ है, तब :

(A) $f(4) - g(4) = 10$ (B) $|f(x) - g(x)| < 2 \Rightarrow -2 < x < 0$

(C) $f(x) = g(x) \Rightarrow x = -1$ (D) $f(x) - g(x) = 2x$ का कोई वास्तविक मूल नहीं रखता है

Ans. ABCD

Sol. Integrating twice $f(x) = g(x) + 2x + 2$

I02/2/038. Let $b > a > 0$, $I_1 = \int_a^{\frac{a+b}{2}} \left(\sqrt{\frac{b-x}{x-a}} - \sqrt{\frac{x-a}{b-x}}\right)dx$ and $I_2 = \int_{\frac{a+b}{2}}^{b} \left(\sqrt{\frac{x-a}{b-x}} - \sqrt{\frac{b-x}{x-a}}\right)dx$, then :

माना $b > a > 0$, $I_1 = \int_a^{\frac{a+b}{2}} \left(\sqrt{\frac{b-x}{x-a}} - \sqrt{\frac{x-a}{b-x}}\right)dx$ और $I_2 = \int_{\frac{a+b}{2}}^{b} \left(\sqrt{\frac{x-a}{b-x}} - \sqrt{\frac{b-x}{x-a}}\right)dx$, तब :

(A) $I_1 = I_2$ (B) $I_1 > I_2$ (C) $I_1 < I_2$ (D) $I_1 + I_2 = 10$ यदि $b - a = 5$

Ans. AD

Sol.

$$I_1 = \int_a^{\frac{a+b}{2}} \frac{(a+b)-2x}{\sqrt{(x-a)(b-x)}} = 2\sqrt{(x-a)(b-x)}\Big|_a^{\frac{a+b}{2}} = (b-a)$$

$$I_2 = \int_{\frac{a+b}{2}}^{b} \frac{2x-(a+b)}{\sqrt{(x-a)(b-x)}}dx = -2\sqrt{(x-a)(b-x)}\Big|_{\frac{a+b}{2}}^{b} = b-a$$

Hence अत: $I_1 = I_2 = b - a$

I02/2/039. The value of $\int_0^{\infty} 4[x+7]e^{-2x}dx$ is $\frac{ae^2-12}{e^2-b}$, then (Here [.] represents greatest integer function) :

$\int_0^{\infty} 4[x+7]e^{-2x}dx$ का मान $\frac{ae^2-12}{e^2-b}$ है, तब (जहाँ [.] महत्तम पूर्णांक फलन है) :

(A) $a = 14$ (B) $a = 13$ (C) $b = 2$ (D) $b = 1$

Ans. AD

Sol. To evaluate the floor function, split the integral into unit intervals

निम्न समाकल को इकाई अंतराल में विभक्त करने पर

$$\int_0^{\infty} 4[x+7]e^{-2x}dx = \sum_{k=0}^{\infty} \int_k^{k+1} 4(k+7)e^{-x}dx$$

$= (14e^{-0} - 14e^{-2}) + (16e^{-2} - 16e^{-4}) + (18e^{-4} - 18e^{-6})$ ....

$= 12 + 2(e^{-0} + e^{-2} + e^{-4} + ...) = 12 + \frac{2}{1-e^{-2}} = \frac{14-12e^{-2}}{1-e^{2}} = \frac{14e^2-12}{e^2-1}$

I02/2/040. The value of the definite integral $\int_{-\infty}^{a} \frac{\left(\sin^{-1} e^{x} + \sec^{-1} e^{-x}\right) dx}{\left(\cot^{-1} e^{a} + \tan^{-1} e^{x}\right)\left(e^{x} + e^{-x}\right)}$ $(a \in R)$ is :

(A) Independent of a (B) dependent on a

(C) $\frac{\pi}{2} \ell n 2$ (D) $-\frac{\pi}{2} \ell n\left(\frac{2}{\pi} \tan^{-1} e^{-a}\right)$

समाकल $\int_{-\infty}^{a} \frac{\left(\sin^{-1} e^{x} + \sec^{-1} e^{-x}\right) dx}{\left(\cot^{-1} e^{a} + \tan^{-1} e^{x}\right)\left(e^{x} + e^{-x}\right)}$ $(a \in R)$ का मान है :

(A) a से स्वतंत्र (B) a पर निर्भर

(C) $\frac{\pi}{2} \ell n 2$ (D) $-\frac{\pi}{2} \ell n\left(\frac{2}{\pi} \tan^{-1} e^{-a}\right)$

Ans. BD

Sol. We have जहाँ

$$I = \int_{-\infty}^{a} \left(\frac{\sin^{-1} e^{x} + \cos^{-1} e^{x}}{\cot^{-1} e^{a} + \tan^{-1} e^{x}}\right)\left(\frac{e^{x}}{e^{2x} + 1}\right) dx = \frac{\pi}{2} \int_{-\infty}^{a} \frac{1}{\left(\cot^{-1} e^{a} + \tan^{-1} e^{x}\right)}\left(\frac{e^{x}}{\left(e^{2x} + 1\right)}\right) dx$$

Put रखने पर $\tan^{-1} e^{x} = t$ $\Rightarrow \frac{e^{x}}{e^{2x} + 1} dx = dt$

$$I = \frac{\pi}{2} \int_{0}^{\tan^{-1} e^{a}} \frac{dt}{\left(t + \cot^{-1} e^{a}\right)} = \frac{\pi}{2}\left[\ell n\left(t + \cot^{-1} e^{a}\right)\right]_{0}^{\tan^{-1} e^{a}}$$

$$= \frac{\pi}{2}\left[\lambda n\left(\frac{\pi}{2}\right) - \ell n\left(\cot^{-1} e^{a}\right)\right] = -\frac{\pi}{2} \ell n\left(\frac{2}{\pi} \tan^{-1} e^{-a}\right)$$

I02/2/041. If $I_1 = \int_{0}^{1} \frac{1 + x^{8}}{1 + x^{4}} dx$ and $I_2 = \int_{0}^{1} \frac{1 + x^{9}}{1 + x^{3}} dx$, then :

यदि $I_1 = \int_{0}^{1} \frac{1 + x^{8}}{1 + x^{4}} dx$ और $I_2 = \int_{0}^{1} \frac{1 + x^{9}}{1 + x^{3}} dx$, तब :

(A) $I_2 < I_1 < \frac{\pi}{4}$ (B) $\frac{\pi}{4} < I_2 < I_1$ (C) $1 < I_1 < I_2$ (D) $I_2 < I_1 < 1$

Ans. BD

Sol. For all $x \in (0, 1)$ के लिए

$$\Rightarrow \frac{1}{1+x^2} < \frac{1+x^9}{1+x^3} < \frac{1+x^8}{1+x^4} < 1 \qquad \therefore \int_0^1 \frac{1}{1+x^2}dx < I_2 < I_1 < \int_0^1 1dx \qquad \therefore \frac{\pi}{4} < I_2 < I_1 < 1$$

I02/2/042. If g(x) = {x}[x], where {.} and [.] represents fractional part and greatest integer function respectively and $f(k) = \int_k^{k+1} g(x)dx$ (k ∈ N), then :

(A) f(1), f(2), f(3), ... are in H.P.

(B) $\sum_{r=1}^{\infty}(-1)^{r+1} f(r) = 1 - \ell n2$

(C) $\sum_{r=1}^{\infty}(-1)^{r} f(r) = \ell n\left(\frac{2}{e}\right)$

(D) $\sum_{r=0}^{n} f\left(\frac{1}{r}\right) = \frac{n(n+1)}{2}$

यदि g(x) = {x}[x], जहाँ {.} और [.] भिन्नात्मक और महत्तम पूर्णांक फलन को दर्शाता है और $f(k) = \int_k^{k+1} g(x)dx$ (k ∈ N), तब :

(A) f(1), f(2), f(3), ... हरात्मक श्रेढ़ी में है

(B) $\sum_{r=1}^{\infty}(-1)^{r+1} f(r) = 1 - \ell n2$

(C) $\sum_{r=1}^{\infty}(-1)^{r} f(r) = \ell n\left(\frac{2}{e}\right)$

(D) $\sum_{r=0}^{n} f\left(\frac{1}{r}\right) = \frac{n(n+1)}{2}$

Ans. ABC

Sol. $f(k) = \int_k^{k+1} (x-k)^k dx = \left(\frac{(x-k)^{k+1}}{k+1}\right)_k^{k+1} \Rightarrow f(k) = \frac{1}{k+1}$

Now अब $\sum_{r=1}^{\infty}(-1)^{r+1} f(r)$ = f(1) – f(2) + f(3) + ...

$= \frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \frac{1}{5} + ... = 1 - \ell n2$

I02/2/043. If $\int_2^x g(t)dt = \frac{x^2}{2} + \int_x^2 t^2 g(t)dt$, then equation g(x) = λ has :

(A) 2 solution if $|\lambda| < \frac{1}{2}$

(B) 2 solution if $|\lambda| < \frac{1}{2}$ & $\lambda \neq 0$

(C) 1 solution if $\lambda = -\frac{1}{2}$ (D) No solution if $|\lambda| > \frac{1}{2}$

यदि $\int_2^x g(t)\,dt = \frac{x^2}{2} + \int_x^2 t^2 g(t)\,dt$, तब समीकरण g(x) = λ रखता है :

(A) 2 हल यदि $|\lambda| < \frac{1}{2}$ (B) 2 हल यदि $|\lambda| < \frac{1}{2}$ & $\lambda \neq 0$

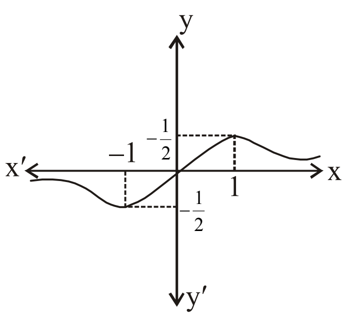
(C) 1 हल यदि $\lambda = -\frac{1}{2}$ (D) कोई हल नहीं यदि $|\lambda| > \frac{1}{2}$

Ans. BCD

Sol. Differentiating both sides दोनों ओर अवकलन करने पर

$g(x) = x - x^2 g(x) \Rightarrow g(x) = \frac{x}{1+x^2}$

Now, graph of y = g(x) is अब y = g(x) का लेखाचित्र है



I02/2/044. $\int_0^1 \frac{x^4\left(1+x^{10065}\right)}{\left(1+x^5\right)^{2015}}dx = \frac{1}{p}$, then :

(A) Number of ways in which p can be expressed as a product of two relatively prime factors is 8.
(B) Number of ways in which p can be expressed as a product of two relatively prime factors is 4.
(C) Number of ways in which p can be expressed as a product of two factors is 8.
(D) Number of ways in which p can be expressed as a product of two factors is 4.

यदि $\int_0^1 \frac{x^4\left(1+x^{10065}\right)}{\left(1+x^5\right)^{2015}}dx = \frac{1}{p}$, तब :

(A) उन तरीकों की संख्या जिनमें p को दो सह–अभाज्य गुणनखण्डों में निरूपित कर सकें, 8 हैं।
(B) उन तरीकों की संख्या जिनमें p को दो सह–अभाज्य गुणनखण्डों में निरूपित कर सके, 4 हैं।
(C) उन तरीकों की संख्या जिनमें p को दो गुणनखण्डों में निरूपित कर सकें, 8 हैं।

(D) उन तरीकों की संख्या जिनमें p को दो गुणनखण्डों में निरूपित कर सकें, 4 हैं।

Ans. AC

Sol. Put $x^5 = t$ रखने पर

$$I = \frac{1}{5}\int_0^1 \frac{1+t^{2013}}{(1+t)^{2015}}\,dt$$

$$= \frac{1}{5}\int_0^1 \frac{1}{(1+t)^{2015}}\,dt + \frac{1}{5}\int_0^1 \frac{t^{-2}}{(t^{-1}+1)^{2015}}\,dt$$

$$= \frac{1}{5}\times\frac{1}{2014}$$

$\therefore\ p = 5 \times 2014 = 2 \times 5 \times 19 \times 53$

I02/2/045. If $I_n = \int_0^{\pi}\left(1-|\sin x|+|\sin 2x|-|\sin 3x|+.....\text{upto n terms}\right)dx$ then :

यदि $I_n = \int_0^{\pi}\left(1-|\sin x|+|\sin 2x|-|\sin 3x|+.....\text{upto n terms}\right)dx$ तब :

(A) $I_{(2)} = \pi - 2$ (B) $I_{(3)} = \pi$ (C) $I_{(2008)} = \pi - 2$ (D) $I_{(2009)} = \pi$

Ans. ABCD

Sol. Let $I = \int_0^{\pi}|\sin \lambda x|\,dx$, $\lambda \in N$

put $\lambda x = t$ $\Rightarrow \lambda dx = dt$

$$\Rightarrow I = \frac{1}{\lambda}\int_0^{\lambda\pi}|\sin t|\,dt = \int_0^{\pi}|\sin t|\,dt = 2$$

Using this

$$I_{(2)} = \int_0^{\pi}\left(1-|\sin x|\right)dx = \pi - 2$$

$$I_{(3)} = \int_0^{\pi}\left(1-|\sin x|+|\sin 2x|\right)dx$$

$$= \int_0^{\pi} 1dx - \int_0^{\pi} |\sin x| dx + \int_0^{\pi} |\sin 2x| dx = \pi - 2 + 2 = \pi$$

Similarly $I_{(2008)} = \pi - 2$ & $I_{(2009)} = \pi$

I02/2/046. If $I_n = \int_0^1 (1-x^2)^n dx$, then :

यदि $I_n = \int_0^1 (1-x^2)^n dx$ तब :

(A) $I_n = \frac{2n}{2n+1} I_n - 1$

(B) $I_n = \frac{2.4.6.......2n}{3.5.7.....(2n+1)}$

(C) $I_n = \frac{2^n n!}{3.5.7.....(2n+1)}$

(D) $I_n = \frac{2n+1}{2n} I_n - 1$

Ans. ABC

Sol.

$$I_n = \int_0^1 (1-x^2)^n dx$$

$$= (1-x^2)^n \cdot x\Big|_0^1 + 2n\int_0^1 (1-x^2)^{n-1} x^2 dx$$

$$= 2n\int_0^1 (1-x^2)^{n-1} \{1-(1-x^2)\} dx$$

$$I_n = 2nI_{n-1} - 2nI_n \qquad \Rightarrow I_n = \frac{2n}{2n+1} I_{n-1}$$

$$I_1 = \int_0^1 (1-x^2) dx = \frac{2}{3}$$

$$I_2 = \frac{4}{5} \cdot \frac{2}{3}, \; I_3 = \frac{6.4.2}{7.5.3}$$

$\vdots$

$$I_n = \frac{2.4.6....2n}{3.5.7.....(2n+1)} = \frac{2^n n!}{3.5.7.....(2n+1)}$$

I02/2/047. Let $f:[0,\infty)\to R$ be a continuous strictly increasing function, such that $f^3(x)=\int_0^x tf^2(t)\,dt\ \forall\ x\ge 0$, then which of the following is/are **INCORRECT** :

(A) f(x) is onto function

(B) $\int_0^1 f(x)\,dx=\frac{1}{18}$

(C) Number of solutions of $6f(x)=e^x$ is 2

(D) graph of $y=6f(x)$ intersects the graph of $y=3x^2+2x+2$ at one point

माना कि $f:[0,\infty)\to R$ एक निरंतर वर्धमान फलन इस प्रकार है कि $f^3(x)=\int_0^x tf^2(t)\,dt\ \forall\ x\ge 0$, तब निम्न में से कौनसा/से सही होगा/होंगे ?

(A) f(x) एक आच्छादक फलन है

(B) $\int_0^1 f(x)\,dx=\frac{1}{18}$

(C) $6f(x)=e^x$ के हलों की संख्या 2 है

(D) वक्र $y=6f(x)$ का आलेख वक्र $y=3x^2+2x+2$ के आलेख को एक बिन्दु पर प्रतिच्छेद करता है

Ans. ACD

Sol. $f^3(x)=\int_0^x tf^2(t)\,dt$

Differentiate both sides

$3f2(x)\,f'(x)=x\,f^2(x)$

$\Rightarrow f'(x)=\frac{x}{3}$ As $x\ge 0$

$\Rightarrow$ f(x) is increasing

$\Rightarrow \boxed{f(x)=\frac{x^2}{6}}$ as $f(0)=0$

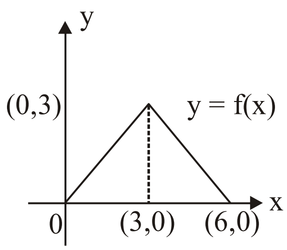
I02/2/048. Let $f(x-3)=f(x+3)$ for all $x\in R$ and $f(x)=\begin{cases} x & ;\ 0\le x<3 \\ 6-x & ;\ 3\le x<6 \end{cases}$, then :

माना $f(x-3)=f(x+3)$ सभी $x\in R$ के लिए तथा $f(x)=\begin{cases} x & ;\ 0\le x<3 \\ 6-x & ;\ 3\le x<6 \end{cases}$, तब :

(A) $\int_0^{36} f(x)dx = 54$ (B) $\int_0^{48} f(x)dx = 64$ (C) $\int_0^{54} f(x)dx = 81$ (D) $\int_0^{18} f(x)dx = 9$

Ans. AC

Sol. f(x) is periodic with period '6'



$$\therefore \int_0^{6} f(x)dx = \frac{1}{2}\times 6\times 3 = 9$$

I02/2/049. Let f(x) is differentiable bijective function such that $f'(x) - \int_0^{x} f(t)dt = \int_1^{f(x)} g(t)dt \ \forall x > 0$, f(0) = 1 and $g(x) = f^{-1}(x)$, then :

माना f(x) एक अवकलनीय एकैकी–आच्छादक फलन इस प्रकार है कि $f'(x) - \int_0^{x} f(t)dt = \int_1^{f(x)} g(t)dt \ \forall x > 0$, f(0) = 1 तथा $g(x) = f^{-1}(x)$, तो :

(A) $g(e) = \sqrt{2}$ (B) $g(1) = \sqrt{e}$ (C) $f(1) = \sqrt{e}$ (D) $f(e) = \sqrt{2}$

Ans. AC

Sol. $\int_0^{x} f(t)dt + \int_1^{f(x)} g(t)dt = f'(x)$

Using : $\int_a^{b} f(x)dx + \int_{c=f(a)}^{d=f(b)} f^{-1}(x)dx = bd - ac$

$\Rightarrow x.f(x) - 0 \times 1 = f'(x)$ (think !)

$\Rightarrow \frac{f'(x)}{f(x)} = x$ $(\because f(x) \neq 0)$

$\Rightarrow \ell n f(x) = \frac{x^2}{2} + C$

put $x = 0 \Rightarrow C = 0$

$\Rightarrow f(x) = e^{\frac{x^2}{2}}$

$f(\sqrt{2}) = e \Rightarrow g(e) = \sqrt{2}$

$f(1) = \sqrt{e}$

I02/2/050. $f : [0, 1] \to R$ is a function with continuous second derivative and $f'(x) \in (0, 1], \forall x \in [0, 1]$. Let $\int_0^x f(t)dt = g(x)$, $f(0) = g(0)$ and $\int_0^x (f(t))^3 dt = h(x)$, then :

$f : [0, 1] \to R$ एक फलन है जिसका द्वितीय अवकलज सतत् है तथा $f'(x) \in (0, 1], \forall x \in [0, 1]$ है। माना $\int_0^x f(t)dt = g(x)$, $f(0) = g(0)$ तथा $\int_0^x (f(t))^3 dt = h(x)$, तब :

(A) $h(x) \le g(x)$ (B) $(g(x))^2 \ge h(x)$

(C) $2g(x) \ge (f(x))^2$ (D) $f(1) = 1$

Ans. ABC

Sol. Let $P(x) = (g(x))^2 - h(x)$

$p'(x) = f(x)\,(2g(x) - (f(x))^2)$

Let $q(x) = 2g(x) - (f(x))^2$

$q'(x) = 2f(x)\,(1 - f'(x)) \ge 0 \ \forall x \in [0, 1]$

$\Rightarrow q(x) \ge 0, \forall x \in [0,1]$

$\Rightarrow p'(x) \ge 0, \forall x \in [0, 1]$

$\Rightarrow p(x) \ge 0 \ \forall x \in [0,1]$

I02/2/051. $f : [0, 1] \to R$ be a non-increasing function, then for $\alpha \in (0,1)$:

$f : [0, 1] \to R$ एक अवर्द्धमान फलन है तब $\alpha \in (0,1)$ के लिए :

(A) $\alpha\int_0^1 f(x)dx \le \int_0^\alpha f(x)dx$ (B) $\alpha\int_0^1 f(x)dx \ge \int_0^\alpha f(x)dx$

(C) $\alpha^2\int_0^1 f(x)dx \le \int_0^\alpha f(x)dx$ (D) $\sqrt{\alpha}\int_0^1 f(x)dx \ge \int_0^\alpha f(x)dx$

Ans. AC

Sol. $\int_0^{\alpha} f(x)\,dx$ put $x = \alpha t$, $dx = \alpha dt$

$$\alpha\int_0^1 f(\alpha t)\,dt \ge \alpha\int_0^1 f(t)\,dt$$

Since $\alpha \in (0, 1)$

I02/2/052. Let $P(x) = a_n x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + ... + a_1 x + a_0$, then $P(0) + P'(0) + P''(0) + ...$ is equal to :

माना $P(x) = a_n x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + ... + a_1 x + a_0$, तब $P(0) + P'(0) + P''(0) + ...$ बराबर है :

(A) $P(1) + P(2) + ... P(n)$

(B) $a_0 + a_1 + 2a_2 + 3!a_3 .... n!a_n$

(C) $\int_0^{\infty} e^{-x}P(x)\,dx$

(D) $\int_0^{-\infty} e^{x}P(x)\,dx$

Ans. BC

Sol. $\int_0^{\infty} e^{-x}P(x)\,dx = -\left|P(x)e^{-x}\right|_0^{\infty} + \int_0^{\infty} e^{-x}P'(x)\,dx$

$$= P(0) - \left|P'(x)e^{-x}\right|_0^{\infty} + \int_0^{\infty} e^{-x}P''(x)\,dx$$

I02/2/053. Let $a + b = 4$ where $a, b > 0$ and $a < 2$ and $f(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^5}{5}$, then :

(A) $\int_0^a f(x)\,dx + \int_0^b f(x)\,dx$ increases as $(b - a)$ increases

(B) $\int_0^a f^{-1}(x)\,dx + \int_0^b f^{-1}(x)\,dx$ increases as $(b - a)$ increases

(C) $\int_0^a \frac{1}{f^{-1}(x)}\,dx + \int_0^b \frac{1}{f^{-1}(x)}\,dx$ decreases as $(b - a)$ increases

(D) $\int_0^a \frac{1}{f(x)}\,dx + \int_0^b \frac{1}{f(x)}\,dx$ decreases as $(b - a)$ increases

माना $a + b = 4$ जहाँ $a, b > 0$ तथा $a < 2$ तथा $f(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^5}{5}$, तब :

(A) $\int_0^a f(x)dx + \int_0^b f(x)dx$ वर्धमान है जब (b – a) वर्धमान है

(B) $\int_0^a f^{-1}(x)dx + \int_0^b f^{-1}(x)dx$ वर्धमान है जब (b – a) वर्धमान है

(C) $\int_0^a \frac{1}{f^{-1}(x)}dx + \int_0^b \frac{1}{f^{-1}(x)}dx$ ह्रासमान है जब (b – a) वर्धमान है

(D) $\int_0^a \frac{1}{f(x)}dx + \int_0^b \frac{1}{f(x)}dx$ ह्रासमान है जब (b – a) वर्धमान है

Ans. ABCD

Sol. $\because f'(x) > 0$; f(x) is an increasing function

Let $b - a = t$

So, $a = \frac{4+t}{2}, b = \frac{4-t}{2}$

Let $Q(t) = \int_0^a f(x)dx + \int_0^b f(x)dx = \int_0^{\frac{4+t}{2}} f(x) + \int_0^{\frac{4-t}{2}} f(x)dx$

Now find $Q'(t)$

I02/2/054. Let $I_1 = \int_0^{\frac{\pi}{4}} (\tan x)^{\frac{1}{3}} dx$ and $\int_0^{\frac{\pi}{4}} (\tan x)^{\sqrt{2}} dx$, then :

माना $I_1 = \int_0^{\frac{\pi}{4}} (\tan x)^{\frac{1}{3}} dx$ तथा $\int_0^{\frac{\pi}{4}} (\tan x)^{\sqrt{2}} dx$, तब :

(A) $I_1 < \frac{3}{4}$ (B) $I_2 < \frac{\pi}{4}$ (C) $2 < I_1 < 3$ (D) $I_2 < \log\sqrt{2} < I_1$

Ans. ABD

Sol. Put $\tan x = t^{\frac{3}{2}}$

So, $I_1 = \frac{3}{2}\int_0^1 \frac{tdt}{t^3+1}$

Now draw the graph of $f(t)=\frac{t}{t^3+1}$ and use estimation

Also draw the graph of $y=(\tan x)^{\frac{1}{3}}$ and $y=(\tan x)^{\sqrt{2}}$

Clearly $I_1 > I_2$

I02/2/055. f(x) is a differentiable function such that $f(x)=x^2+\int_0^x e^{-t}f(x-t)\,dt$, then :

f(x) एक अवकलनीय फलन इस प्रकार है कि $f(x)=x^2+\int_0^x e^{-t}f(x-t)\,dt$, तब :

(A) $f(x)=\frac{x^3}{3}+x$ (B) $f(x)=\frac{x^3}{3}+x^2$ (C) $f'(x)=x^2+2x$ (D) $\int_{-2}^{2}((x)-x^2)\,dx=0$

Ans. BCD

Sol. $f(x)=x^2+\int_0^x e^{-t}f(x-t)\,dt$

$=x^2+\int_0^x e^{-(x-t)}.f(x-(x-t))\,dt$

$=x^2+e^{-x}\cdot\int_0^x e^{t}\cdot f(t)\,dt$

$\therefore e^{x}\cdot f(x)=x^2\cdot e^{x}+\int_0^x e^{t}\cdot f(t)\,dt$

Differentiate with respect to x, we get $f'(x)=x^2+2x$

$\Rightarrow f(x)=\frac{x^3}{3}+x^2+c$, also $f(0)=0$

$\Rightarrow c=0$

$\Rightarrow f(x)=\frac{x^3}{3}+x^2$

I02/2/056. If $I_1=\int_0^{\frac{\pi}{2}}\frac{\sin(\sin x)}{\sin x}dx$; $I_2=\int_0^{\frac{\pi}{2}}\frac{\sin x}{x}dx$; $I_3=\int_0^{\frac{\pi}{2}}\frac{\sin(\tan x)}{\tan x}dx$, then which of the following is true :

यदि $I_1 = \int_0^{\frac{\pi}{2}} \frac{\sin(\sin x)}{\sin x} dx$; $I_2 = \int_0^{\frac{\pi}{2}} \frac{\sin x}{x} dx$; $I_3 = \int_0^{\frac{\pi}{2}} \frac{\sin(\tan x)}{\tan x} dx$, तब निम्न में से कौनसा/से सत्य होगा/होंगे :

(A) $I_1 > I_3$ (B) $I_2 > I_3$ (C) $I_1 > I_2$ (D) $I_1 < I_2$

Ans. ABC

Sol. $0 < x < \frac{\pi}{2} \Rightarrow \frac{\sin x}{x}$ is decreasing and $\sin x < x < \tan x$

$$\Rightarrow \frac{\sin(\sin x)}{\sin x} > \frac{\sin x}{x} > \frac{\sin(\tan x)}{\tan x}$$

$\Rightarrow I_1 > I_2 > I_3$

I02/2/057. $I = \int_1^e \ell n x e^x dx$:

(A) $I > \frac{e^{e+1}}{2+2e}$ (B) $I < \frac{e^{e+1}}{2+2e}$ (C) $I > \frac{e^{e+1} - e^e}{2}$ (D) $I < \frac{e^{e+1} - e^e}{2}$

Ans. AD

Sol. Let $y = \ell n\, x\, e^x$

$$y' = \left(\frac{1}{x} + \ell nx\right)e^x$$

$$y'' = \left(\frac{2}{x} - \frac{1}{x^2} + \ell nx\right)e^x$$

$y' > 0\ \forall\ x \in (1, e)$

$\Rightarrow$ Area of Δ (1, 0), (e, e^e), (e, 0) is greater than I which is $\frac{1}{2}(e-1)e^e$

Also slope of tangent at e, e^e is $\frac{1+e}{e} \cdot e^e$

Area formed by tangent is less than I

I02/2/058. $\int_0^{\frac{\pi}{2}} \frac{\sin^6 x}{\sin x + \cos x} dx$ is equal to :

$$\int_0^{\frac{\pi}{2}} \frac{\sin^6 x}{\sin x + \cos x} dx$$ बराबर है :

(A) $\frac{1}{4\sqrt{2}} \log\left(\sqrt{2}+1\right) + \frac{1}{4}$

(B) $-\frac{1}{4\sqrt{2}} \log\left(\sqrt{2}-1\right) + \frac{1}{4}$

(C) $\frac{1}{4\sqrt{2}} \log\left(\sqrt{2}+1\right) - \frac{1}{4}$

(D) $-\frac{1}{4\sqrt{2}} \log\left(\sqrt{2}-1\right) - \frac{1}{4}$

Ans. AB

Sol. Let $I = \int_0^{\frac{\pi}{2}} \frac{\sin^6 x}{\sin x + \cos x} dx$

$$I = \int_0^{\frac{\pi}{2}} \frac{\cos^6 x}{\sin x + \cos x} dx \quad \left(\int_0^a f(x) dx = \int_0^a f(a-x) dx \right)$$

$$2I = \int_0^{\frac{\pi}{2}} \frac{\sin^{6x} + \cos^6 x}{\sin x + \cos x} dx$$

$$I = \int_0^{\frac{\pi}{4}} \frac{\sin^6 x + \cos^6 x}{\sin x + \cos x} dx \quad \left(\int_0^a f(x) dx = 2\int_0^{\frac{a}{2}} f(x) dx \right) \quad (\text{if } f(a-x) = f(x))$$

$$I = \int_0^{\frac{\pi}{4}} \frac{1 - 3\sin^2 x \cos^2 x}{\sin x + \cos x} dx = \frac{1}{4}\int_0^{\frac{\pi}{4}} \frac{4 - 3\sin^2 2x}{\sin x + \cos x} dx$$

$$I = \frac{1}{4}\int_0^{\frac{\pi}{4}} \frac{4 - 3\sin^2\left(2\left(\frac{\pi}{4} - x\right)\right)}{\sin\left(\frac{\pi}{4} - x\right) + \cos\left(\frac{\pi}{4} - x\right)} dx \quad \left(\int_0^a f(x) dx = \int_0^a f(a-x) dx \right)$$

$$I = \frac{1}{4\sqrt{2}}\int_0^{\frac{\pi}{4}} \frac{4 - 3\cos^2 2x}{\cos x} dx$$

$$I = \frac{1}{4\sqrt{2}}\int_0^{\frac{\pi}{4}} \frac{4 - 3\left(2\cos^2 x - 1\right)^2}{\cos x} dx$$

$$I=\frac{1}{4\sqrt{2}}\int_0^{\frac{\pi}{4}}\left(\sec x+12\cos x-12\cos^3 x\right)dx$$

$$I=\frac{1}{4\sqrt{2}}\left[\ell n\left(\sec x+\tan x\right)+12\sin x-12\left(\sin x-\frac{\sin^3 x}{3}\right)\right]_0^{\frac{\pi}{4}}=\frac{1}{4\sqrt{2}}\left[\ell n\left(\sqrt{2}+1\right)\right]+\frac{1}{4}$$

I02/2/059. If $|a|<1$; if $\int_0^{\pi}\frac{\ell n\left(1+a\cos x\right)}{\cos x}dx=\pi f\left(a\right)$ then :

यदि $|a|<1$; यदि $\int_0^{\pi}\frac{\ell n\left(1+a\cos x\right)}{\cos x}dx=\pi f\left(a\right)$ तब :

(A) $f(x)=\sin^{-1}x$ (B) $f\left(1\right)=\frac{\pi}{2}$ (C) $f(x)=\cos^{-1}x$ (D) $f(1)=0$

Ans. AB

Sol. Let $I\left(a\right)=\int_0^{\pi}\frac{\ell n\left(1+a\cos x\right)}{\cos x}dx$ $(I(0)=0)$

Differentiate w.r.t. 'a'

$$I'\left(a\right)=\int_0^{\pi}\frac{1}{\left(1+a\cos x\right)}\frac{\cos x}{\cos x}dx=\int_0^{\pi}\frac{dx}{1+a\cos x}=\int_0^{\pi}\frac{dx}{1+a\cos\left(\pi-x\right)}=\int_0^{\pi}\frac{dx}{1-a\cos x}$$

$$2I'\left(a\right)=\int_0^{\pi}\frac{2dx}{\left(1-a^2\cos^2 x\right)}$$

$$I'\left(a\right)=2\int_0^{\frac{\pi}{2}}\frac{\sec^2 xdx}{\left(\sec^2 x-a^2\right)};\qquad (\tan x=t;\ \sec^2xdx=dt)$$

$$I'\left(a\right)=2\int_0^{\infty}\frac{dt}{t^2+\left(\sqrt{1-a^2}\right)^2}=\frac{2}{\sqrt{1-a^2}}\left[\tan^{-1}\frac{t}{\sqrt{1-a^2}}\right]_0^{\infty}=\frac{2}{\sqrt{1-a^2}}\left[\frac{\pi}{2}-0\right]=\frac{\pi}{\sqrt{1-a^2}}$$

$$I'\left(a\right)=\frac{\pi}{\sqrt{1-a^2}}$$

$$I\left(a\right)=\int\frac{\pi}{\sqrt{1-a^2}}da$$

$I(a) = \pi \sin^{-1}a + c \ (c = 0, I(0) = 0)$

$I(a) = \pi \sin^{-1}a = \pi f(a)$

$f(a) = \sin^{-1}(a)$

$f(x) = \sin^{-1}(x)$

$f(1) = \frac{\pi}{2}$

I02/2/060. If $f(x) = -\int_0^x \ell n(\cos t)\,dt$, then the value of $f(x) - 2f\left(\frac{\pi}{4}+\frac{x}{2}\right) + 2f\left(\frac{\pi}{4}-\frac{x}{2}\right)$, is equal to :

यदि $f(x) = -\int_0^x \ell n(\cos t)\,dt$, तब $f(x) - 2f\left(\frac{\pi}{4}+\frac{x}{2}\right) + 2f\left(\frac{\pi}{4}-\frac{x}{2}\right)$ का मान बराबर होगा :

(A) $-x\ell n2$ (B) $\frac{x}{2}\ell n2$ (C) $\frac{x}{3}\ell n3$ (D) $x\ell n\left(\frac{1}{2}\right)$

Ans. AD

Sol.

$$f\left(\frac{\pi}{4}+\frac{x}{2}\right) = -\int_0^{\frac{\pi}{4}} \ell n(\cos t)\,dt - \int_{\frac{\pi}{4}}^{\frac{\pi}{4}+\frac{x}{2}} \ell n(\cos t)\,dt$$

$$f\left(\frac{\pi}{4}-\frac{x}{2}\right) = -\int_0^{\frac{\pi}{4}} \ell n(\cos t)\,dt - \int_{\frac{\pi}{4}}^{\frac{\pi}{4}-\frac{x}{2}} \ell n(\cos t)\,dt$$

$$2f\left(\frac{\pi}{4}+\frac{x}{2}\right) - 2f\left(\frac{\pi}{4}-\frac{x}{2}\right) = 2\int_{\frac{\pi}{4}}^{\frac{\pi}{4}-\frac{x}{2}} \ell n(\cos t)\,dt - 2\int_{\frac{\pi}{4}}^{\frac{\pi}{4}+\frac{x}{2}} \ell n(\cos t)\,dt$$

$$= -2\int_0^{\frac{x}{2}} \ell n\cos\left(\frac{\pi}{4}-t\right)dt - 2\int_0^{\frac{x}{2}} \ell n\cos\left(t+\frac{\pi}{4}\right)dt = -2\int_0^{\frac{x}{2}} \ell n\left(\frac{\cos^2 t - \sin^2 t}{2}\right)dt$$

$$= x\ell n2 - 2\int_0^{\frac{x}{2}} \ell n(\cos 2t)\,dt = x\ell n2 - \frac{2}{2}\int_0^{x} \ell n(\cos t)\,dt = x\ell n2 + f(x)$$

So, $$f(x) - 2f\left(\frac{\pi}{4}+\frac{x}{2}\right) + 2f\left(\frac{\pi}{4}-\frac{x}{2}\right) = -x\ell n2$$

I02/2/061. If $I_1 = \int_0^{\infty} f\left(x^n + x^{-n}\right)\ell nx \cdot \frac{dx}{x}$ & $I_2 = \int_0^{\infty} f\left(x^n + x^{-n}\right)\ell nx \cdot \frac{dx}{\left(1+x^2\right)}$ then :

यदि $I_1 = \int_0^{\infty} f\left(x^n + x^{-n}\right)\ell nx \cdot \frac{dx}{x}$ तथा $I_2 = \int_0^{\infty} f\left(x^n + x^{-n}\right)\ell nx \cdot \frac{dx}{\left(1+x^2\right)}$ तब :

(A) $I_1 = 0$ (B) $I_2 \neq 0$ (C) $I_1 + I_2 = 0$ (D) $2I_1 + 3I_2 = 0$

Ans. ACD

Sol. Let $t = \ell n\, x \Rightarrow x = e^t$

$dx = e^t dt$

$$I_1 = \int_0^{\infty} f\left(x^n + x^{-n}\right)\left(\ell nx\right)\frac{dx}{x} = \int_{-\infty}^{\infty} \underbrace{f\left(e^{nt} + e^{-nt}\right)t}_{\text{odd function}}\, dt = 0$$

$$\text{Now; } I_2 = \int_0^{\infty} f\left(x^n + x^{-n}\right)\left(\ell nx\right)\frac{dx}{1+x^2} = \int_{-\infty}^{\infty} \underbrace{f\left(e^{nt} + e^{-nt}\right)t + \frac{e^t}{1+e^{2t}}}_{\text{odd function}}\, dt = 0$$

I03/2/062. Let f(x) be a every where differentiable function satisfying $(x-y)\, f(x+y) - (x+y)\, f(x-y) = 4xy(x^2 - y^2)$ $(x, y \in R)$. Given $f(1) = 1$ define a function g(x) as $g(x) = f(x) - 3\left[f(x)\right]^{\frac{1}{3}} + 1$, then which of the following holds true :

(A) $g(x) = 0$ has no solution

(B) $g'(x) = 0$ has two real and distinct roots

(C) $g(\alpha).g(\beta) < 0$; α, β being the roots of $g'(x) = 0$

(D) $g(x) = 0$ has no vertical tangent

माना एक अवकलनीय f(x) समीकरण $(x-y)\, f(x+y) - (x+y)\, f(x-y) = 4xy(x^2 - y^2)$ $(x, y \in R)$ और दिया गया है $f(1) = 1$ तथा $g(x) = f(x) - 3\left[f(x)\right]^{\frac{1}{3}} + 1$ इस तरह से परिभाषित है तो निम्न में से कौनसा सत्य है :

(A) $g(x) = 0$ का कोई हल नहीं है

(B) $g'(x) = 0$ दो वास्तविक व भिन्न मूल रखता है

(C) $g(\alpha).g(\beta) < 0$; $g'(x) = 0$ के मूल α, β हैं

(D) $g(x) = 0$ कोई उर्ध्वाधर स्पर्श रेखा नहीं रखता है

Ans. BCD

Sol. $(x-y)\, f(x+y) - (x+y)\, f(x-y) = 4xy(x^2 - y^2)$

$$\Rightarrow \frac{f(x+y)}{x+y} - \frac{f(x-y)}{x-y} = 4xy$$

So, (इसलिए), $\frac{f(x+h)}{x+h} - \frac{f(x-h)}{x-h} = 4xh$

$$\Rightarrow \left[\frac{f(x+h)-f(x)+f(x)}{x+h}\right] - \left[\frac{f(x-h)-f(x)+f(x)}{x-h}\right] = 4xh$$

$$\Rightarrow \left[\frac{f(x+h)-f(x)}{x+h}\right] - \left[\frac{f(x-h)-f(x)}{x-h}\right]$$

$$= 4xh + f(x)\left[\frac{1}{x-h} - \frac{1}{x+h}\right]$$

$$\Rightarrow \left[\frac{f(x+h)-f(x)}{x+h}\right] - \left[\frac{f(x-h)-f(x)}{x-h}\right]$$

$$= 4xh + \frac{2hf(x)}{x^2-h^2}$$

$$\Rightarrow \left[\frac{f(x+h)-f(x)}{h}\right] \cdot \left(\frac{1}{x+h}\right) + \left[\frac{f(x-h)-f(x)}{-h}\right] \cdot \left(\frac{1}{x-h}\right)$$

$$= 4x + \frac{2f(x)}{x^2-h^2}$$

Taking limit on both sides as $h \to 0$

दोनों तरफ सीमा लेने पर $h \to 0$

$$\Rightarrow \frac{f'(x)}{x} + \frac{f'(x)}{x} = 4x + \frac{2f(x)}{x^2} \Rightarrow \frac{f'(x)}{x} = 2x + \frac{f(x)}{x} \Rightarrow f'(x) + \left[\frac{-f(x)}{x}\right] = 2x^2$$

which is a linear differential equation integrating factor $= e^{-\int \frac{1}{x}dx} = \frac{1}{x}$

जो कि रैखिक अवकल समीकरण है जिसका समाकल गुणक $= e^{-\int \frac{1}{x}dx} = \frac{1}{x}$

$$\Rightarrow f(x) \cdot \frac{1}{x} = \int\left(2x^2 \cdot \frac{1}{x} + c\right)dx \Rightarrow \frac{f(x)}{x} = x^2 + c \Rightarrow f(x) = x^3 + cx$$

I03/2/063. The portion of tangent to a curve intercepted between lines $y = x$ & $y = -x$ is bisected by the point of tangency. If the curve passes through the point (2, 1) then :

(A) eccentricity of the curve is $\sqrt{2}$

(B) eccentricity of the curve is $\frac{1}{\sqrt{2}}$

(C) sum of focal distance of any point on the curve is $2\sqrt{3}$

(D) difference of focal distance of any point on the curve is $2\sqrt{3}$

एक वक्र की किसी स्पर्श रेखा का रेखाओं $y = x$ तथा $y = -x$ के मध्य अन्तःखण्डित भाग स्पर्श बिन्दु द्वारा समद्विभाजित होता है। यदि वक्र बिन्दु (2, 1) से गुजरता है, तो :

(A) वृत्त की उत्केन्द्रता $\sqrt{2}$ है

(B) वृत्त की उत्केन्द्रता $\frac{1}{\sqrt{2}}$ है

(C) वक्र पर स्थित किसी बिन्दु की नाभिय दूरीयों का योग $2\sqrt{3}$ है

(D) वक्र पर स्थित किसी बिन्दु की नाभिय दूरीयों का अंतर $2\sqrt{3}$ है

Ans. AD

Sol.

I03/2/064. Solution of the differential equation $\frac{dy}{dx} = \frac{y^2 - 2xy - x^2}{y^2 + 2xy - x^2}$ is :

(A) $x^2 - y^2 + c(x - y) = 0$

(B) $x^2 + y^2 + c(x + y) = 0$

(C) a straight line if it passes through (1, –1)

(D) a circle if it passes through (1,1)

अवकल समीकरण $\frac{dy}{dx} = \frac{y^2 - 2xy - x^2}{y^2 + 2xy - x^2}$ का हल है :

(A) $x^2 - y^2 + c(x - y) = 0$

(B) $x^2 + y^2 + c(x + y) = 0$

(C) (1, –1) से गुजरने वाली सरल रेखा

(D) (1,1) गामी वृत्त

Ans. BCD

Sol.

I03/2/065. The integral curve of the equation $\left(1 - x^2\right)\frac{dy}{dx} + xy = x$ is :

(A) a conic whose centre is (0, 1)
(B) a conic, length of whose one axis is 2
(C) an ellipse if $|x| < 1$
(D) a hyperbola if $|x| > 1$

समीकरण $\left(1-x^2\right)\frac{dy}{dx}+xy = x$ के समाकलनीय वक्र हैं :
(A) शांकव जिसका केन्द्र (0, 1) है
(B) शांकव जिसके एक अक्ष की लम्बाई 2 है
(C) एक दीर्घवृत्त यदि $|x| < 1$
(D) एक अतिपरवलय यदि $|x| > 1$

Ans. ABCD
Sol.

I03/2/066. If $y = y(x)$ is the solution of differential equation $\frac{dy}{dx} - y = -e^x \tan^2 x$; $y(0) = 0$ then :

(A) $y > 0$ if $x \in \left(0, \frac{\pi}{2}\right)$

(B) $y < 0$ if $x \in \left(0, \frac{\pi}{2}\right)$

(C) number of points of inflection for $y = y(x)$ in is 0 $x \in \left(0, \frac{\pi}{2}\right)$

(D) tangent drawn to the graph of $y = y(x)$ at $(x, y(x))$ where $x \in \left(0, \frac{\pi}{2}\right)$ lies below the graph of $y = y(x)$

यदि $y = y(x)$ अवकल समीकरण $\frac{dy}{dx} - y = -e^x \tan^2 x$; $y(0) = 0$ का हल है, तब :

(A) $y > 0$ यदि $x \in \left(0, \frac{\pi}{2}\right)$

(B) $y < 0$ यदि $x \in \left(0, \frac{\pi}{2}\right)$

(C) 0 $x \in \left(0, \frac{\pi}{2}\right)$ में $y = y(x)$ के लिए नति परिवर्तन बिन्दुओं की संख्या 0 है

(D) $y = y(x)$ आरेख पर बिन्दु $(x, y(x))$ पर स्पर्श रेखा जहाँ $y = y(X)$ के आरेख के नीचे स्थित है जहाँ $x \in \left(0, \frac{\pi}{2}\right)$

Ans. BC

Sol.

I03/2/067. Let $y = f(x)$ be a real valued function satisfying $x\frac{dy}{dx} = x^2 + y - 2$, $f(1) = 1$, then:

(A) $f(x)$ is minimum at $x = 1$
(B) $f(x)$ is maximum at $x = 1$
(C) $f(3) = 5$
(D) $f(2) = 3$

माना $y = f(x)$ एक वास्तविक मान फलन है जो $x\frac{dy}{dx} = x^2 + y - 2$, $f(1) = 1$ को संतुष्ट करता है, तब :

(A) $x = 1$ पर $f(x)$ न्यूनतम है
(B) $x = 1$ पर $f(x)$ अधिकतम है
(C) $f(3) = 5$
(D) $f(2) = 3$

Ans. AC

Sol.

$$\frac{xdy - ydx}{x^2} = \frac{x^2 - 2}{x^2}dx$$

$$\int d\left(\frac{y}{x}\right) = \int\left(1 - \frac{2}{x^2}\right)dx$$

$$y = x^2 - 2x + 2 \qquad (\because f(1) = 1)$$

I03/2/068. If a differentiable function satisfies $(x - y)\, f(x + y) - (x + y)\, f(x - y) = 2(x^2y - y^3)\ \forall\, x, y \in R$ and $f(1) = 2$, then :

(A) $f(x)$ must be polynomial function
(B) $f(3) = 12$
(C) $f(0) = 0$
(D) $f(3) = 13$

यदि एक अवकलनीय फलन $(x - y)\, f(x + y) - (x + y)\, f(x - y) = 2(x^2y - y^3)\ \forall\, x, y \in R$ को संतुष्ट करता है तथा $f(1) = 2$, तब :

(A) $f(x)$ एक बहुपद फलन होगा
(B) $f(3) = 12$
(C) $f(0) = 0$
(D) $f(3) = 13$

Ans. ABC

Sol. Put $y = h$

$$\Rightarrow \quad x[f(x + h) - f(x - h)] - h[f(x + h) + f(x - h)] = 2(x^2h - h^3)]$$

or $$\lim_{h\to 0} x\frac{[f(x+h) - f(x-h)]}{h} - [f(x + h) + f(x - h)] = \lim_{h\to 0} 2(x^2 - h^2)$$

$$\Rightarrow \quad xf'(x) - f(x) = x^2 \qquad \Rightarrow \quad f(x) = x^2 + x$$

I03/2/069. Let $y = f(x)$ satisfies the differential equation $(1+x^2)\frac{dy}{dx} + 2xy = 3x$ and $f(0) = 2$. Then correct statement regarding $y = f(x)$ is/are :

(A) it has exactly one integer in its range
(B) it has exactly two points of inflection
(C) it has only one point of inflection
(D) f(x) attains its minimum value

माना $y = f(x)$ अवकलनीय समीकरण $(1 + x^2)\frac{dy}{dx} + 2xy = 3x$ को सन्तुष्ट करता है तथा $f(0) = 2$. तो निम्न में से कौनसा/कौनसे कथन $y = f(x)$ के लिये सत्य है :

(A) इसके परिसर में ठीक एक पूर्णांक संख्या होगी
(B) इसके ठीक दो नति परिवर्तन बिन्दु है
(C) इसके ठीक एक नति परिवर्तन बिन्दु है
(D) f(x) का न्यूनतम मान विद्यमान है

Ans. AB

Sol. $\frac{dy}{dx} + \frac{2xy}{1+x^2} = \frac{3x}{1+x^2}$

I.F. $= e^{\int \frac{2x}{1+x^2}dx} = 1 + x^2$

$y(1+x^2) = \frac{3x^2}{2} + 2 \qquad (\because f(0) = 2)$

$y = \frac{3x^2}{2(1+x^2)} + \frac{2}{(1+x^2)} \Rightarrow y = \frac{3x^2+4}{2(1+x^2)}$

$\Rightarrow y = 1 + \frac{x^2+2}{2(1+x^2)}$

$\Rightarrow y \ge 1$

I01/2/070. $\int \frac{x^2-1}{x\sqrt{x^4+4x^3+22x^2+4x+1}}dx = a\ell n\left\{(x+1)^2 + \sqrt{x^4+4x^3+22x^2+4x+1}\right\} - b\ell nx + c$, where c is a constant of integration, then :

$\int \frac{x^2-1}{x\sqrt{x^4+4x^3+22x^2+4x+1}}dx = a\ell n\left\{(x+1)^2 + \sqrt{x^4+4x^3+22x^2+4x+1}\right\} - b\ell nx + c$, जहाँ c समाकलन अचर है तब :

(A) $a = 1$ (B) $a = -1$ (C) $b = 1$ (D) $b = -1$

Ans. AC

Sol. $$I=\int\frac{\left(1-\frac{1}{x^2}\right)}{\sqrt{\left(x+\frac{1}{x}\right)^2+4\left(x+\frac{1}{x}\right)+20}}dx=\int\frac{dt}{(t+2)^2+(4)^2}$$

$$=\ell n\left|\left(x+\frac{1}{x}+2\right)+\sqrt{\left(x+\frac{1}{x}\right)^2+4\left(x+\frac{1}{x}\right)^2+20}\right|$$

$$=\ell n\left|(x+1)^2+\sqrt{x^4+4x^3+22x^2+4x+1}\right|-\ell nx+c$$

I01/2/071. $\int\sqrt{\tan^2 x+2}dx=a\tan^{-1}\left(\frac{\tan x}{\sqrt{2+\tan^2 x}}\right)+\frac{1}{2}\ell n\left(\frac{\sqrt{f(x)}+\tan x}{\sqrt{f(x)}-\tan x}\right)+C$, where C is constant of integration

:

(A) $a=1$ (B) $a=2$

(C) Range of f(x) is $[2,\infty)$ (D) Range of f(x) is $[3,\infty)$

$\int\sqrt{\tan^2 x+2}dx=a\tan^{-1}\left(\frac{\tan x}{\sqrt{2+\tan^2 x}}\right)+\frac{1}{2}\ell n\left(\frac{\sqrt{f(x)}+\tan x}{\sqrt{f(x)}-\tan x}\right)+C$, जहाँ c समाकल नियतांक है तब :

(A) $a=1$ (B) $a=2$

(C) f(x) का परिसर $[2,\infty)$ है (D) f(x) का परिसर $[3,\infty)$ है

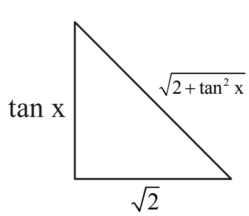
Ans. AC

Sol. $\tan x=\sqrt{2}\tan\phi$

$\sec^2 xdx=\sqrt{2}\sec^2\phi d\phi$

$$dx=\frac{\sqrt{2}\sec^2\phi}{1+2\tan^2\phi}d\phi$$

$$\int\frac{2\sec^3\phi}{1+2\tan^2\phi}d\phi \qquad \sin\phi=y$$



$$\int \frac{2dy}{(1+y^2)(1-y^2)}$$

$$\int \frac{(1+y^2)+(1-y^2)}{(1+y^2)(1-y^2)}dy$$

$$\int \frac{dy}{1+y^2} - \int \frac{dy}{y^2-1}$$

$$= \tan^{-1} y - \frac{1}{2}\ell n\left|\frac{y-1}{y+1}\right| + C$$

$$= \tan^{-1}(\sin\phi) - \frac{1}{2}\ell n\left|\frac{\sin\phi - 1}{y+1}\right| + C$$

$$= \tan^{-1}\left(\frac{\tan x}{\sqrt{2+\tan^2 x}}\right) + \frac{1}{2}\ell n\left(\frac{\sqrt{2+\tan^2 x}+\tan x}{\sqrt{2+\tan^2 x}-\tan x}\right) + C$$

I01/2/072. If $\int \sqrt{\sec^2 x + 3}dx = \ell n\left|f(x)+\sqrt{4-f^2(x)}\right| + \sqrt{3}\sin^{-1}\left(\frac{k.g(x)}{2}\right) + C$, where C is an integration constant, then :

(A) value of k is equal to $\sqrt{3}$ (B) $g(x) = \frac{f(x)}{\sqrt{1+f^2(x)}}$

(C) $g^2(x) + \left(\sqrt{1+f^2(x)}\right)^{-2} = 1$ (D) $g^2(x)\, f^2(x) = f^2(x) - g^2(x)$

यदि $\int \sqrt{\sec^2 x + 3}dx = \ell n\left|f(x)+\sqrt{4-f^2(x)}\right| + \sqrt{3}\sin^{-1}\left(\frac{k.g(x)}{2}\right) + C$, जहाँ C एक समाकलन नियतांक है, तो :

(A) $k = \sqrt{3}$ (B) $g(x) = \frac{f(x)}{\sqrt{1+f^2(x)}}$

(C) $g^2(x) + \left(\sqrt{1+f^2(x)}\right)^{-2} = 1$ (D) $g^2(x)\, f^2(x) = f^2(x) - g^2(x)$

Ans. ACD

Sol. $\int \frac{\sec^2 x+3}{\sqrt{\sec^2 x+3}}dx = \int \frac{\sec^2 x dx}{\sqrt{4+\tan^2 x}} + \int \frac{3\cos x}{\sqrt{4-3\sin^2 x}}dx$

$= \int \frac{dt}{\sqrt{4+t^2}} + \int \frac{\sqrt{3}d\theta}{\sqrt{4-\theta^2}}$

$= \ell n\left|t+\sqrt{4+t^2}\right| + \sqrt{3}\sin^{-1}\left(\frac{\theta}{2}\right)+C$

$= \ell n\left|\tan x+\sqrt{4+\tan^2 x}\right| + \sqrt{3}\sin^{-1}\left(\frac{\sqrt{3}\sin x}{2}\right)+C$

$f(x)=\tan x,\ g(x)=\sin x,\ k=\sqrt{3}$

I01/2/073. If $\int\left(2x^6+15x^4+2x^2+3\right)\cos\left(2x\right)dx$ = f(x) sin (2x) + g(x) cos (2x) + K, where f(x) and g(x) are polynomial of x such that f(0) = 1, g(–1) = –4 and K is constant of integration, then :

यदि $\int\left(2x^6+15x^4+2x^2+3\right)\cos\left(2x\right)dx$ = f(x) sin (2x) + g(x) cos (2x) + K, जहाँ f(x) तथा g(x), x के बहुपद इस प्रकार हैं कि f(0) = 1, g(–1) = –4 तथा K समाकलन नियतांक है, तब :

(A) $4f(1)\le 3g(1)$ (B) $\int_{\frac{-\pi}{3}}^{\frac{\pi}{3}} f\left(x\right)dx \ne 0$ (C) $g''(0)<10$ (D) $\frac{g\left(2\right)}{f\left(2\right)}>1$

Ans. ABCD

Sol. Let $I=\int\left(2x^6+15x^4+2x^2+3\right)\cos 2x dx$

$=\int\left(2x^6+2x^2\right).\cos 2x dx+\int\left(15x^4+3\right).\cos 2x dx$

$\therefore$ Apply integrate by part, we get

$= (x^6 + x^2 + 1)\sin(2x) + (3x^5 + x).\cos(2x) + k$

$\therefore\ f(x) = (x^6 + x^2 + 1);\ g(x) = (3x^5 + x)$

I01/2/074. The value of $\int \frac{dx}{1+e\cos x}$ must be same as :

(A) $\frac{1}{\sqrt{1-e^2}}\tan^{-1}\left(\sqrt{\frac{1-e}{1+e}}\tan\frac{x}{2}\right)+c$ (e lies between 0 and 1)

(B) $\frac{2}{\sqrt{1-e^2}}\tan^{-1}\left(\sqrt{\frac{1-e}{1+e}}\tan\frac{x}{2}\right)+c$, (e is greater than 1)

(C) $\frac{1}{\sqrt{e^2-1}}\log\frac{e+\cos x+\sqrt{e^2-1}\sin x}{1+e\cos x}+c$, (e is greater than 1)

(D) $\frac{2}{\sqrt{e^2-1}}\log\frac{e+\cos x+\sqrt{e^2-1}\sin x}{1+e\cos x}+c$, (e is greater than 1)

$\int\frac{dx}{1+e\cos x}$ का मान किसके बराबर होगा :

(A) $\frac{1}{\sqrt{1-e^2}}\tan^{-1}\left(\sqrt{\frac{1-e}{1+e}}\tan\frac{x}{2}\right)+c$ (e, 0 तथा के मध्य विद्यमान है1)

(B) $\frac{2}{\sqrt{1-e^2}}\tan^{-1}\left(\sqrt{\frac{1-e}{1+e}}\tan\frac{x}{2}\right)+c$, (e, 1 से बड़ा है)

(C) $\frac{1}{\sqrt{e^2-1}}\log\frac{e+\cos x+\sqrt{e^2-1}\sin x}{1+e\cos x}+c$, (e, 1 से बड़ा है)

(D) $\frac{2}{\sqrt{e^2-1}}\log\frac{e+\cos x+\sqrt{e^2-1}\sin x}{1+e\cos x}+c$, (e, 1 से बड़ा है)

Ans. AC

Sol. $I=\frac{2}{1-e}\int\frac{dt}{t^2+\left(\frac{1+e}{1-e}\right)}\quad\left(t=\tan\frac{x}{2}\right)$

If $0<e<1$, $\frac{1+e}{1-e}>0$, So, (B) is correct.

If $e>1$, $\frac{1+e}{1-e}<0$, So, (C) is correct

I01/2/075. Let $f(x)=\frac{4+5\cos x}{5+4\cos x}\left(x\in\left(0,\frac{\pi}{2}\right)\right)$ and $\int\frac{dx}{(5+4\cos x)^2}=\frac{5}{27}g(x)-\frac{4}{27}h(x)+c$ then :

माना $f(x)=\frac{4+5\cos x}{5+4\cos x}\left(x\in\left(0,\frac{\pi}{2}\right)\right)$ तथा $\int\frac{dx}{(5+4\cos x)^2}=\frac{5}{27}g(x)-\frac{4}{27}h(x)+c$ तब :

(A) $g(x)=\tan^{-1}(f(x))$

(B) $h(x)=\sqrt{1+f^2(x)}$

(C) $g(x)=\cos^{-1}(f(x))$ (D) $h(x)=\sqrt{1-f^2(x)}$

Ans. CD

Sol. Let $y=\dfrac{4+5\cos x}{5+4\cos x}\Rightarrow dy=\dfrac{-9\sin x}{(5+4\cos x)^2}dx\Rightarrow \cos x=\dfrac{5y-4}{5-4y}$

$$\therefore \int\frac{dx}{(5+4\cos x)^2}=\frac{-1}{9}\int\frac{dy}{\sin x}=\frac{-1}{9}\int\frac{dy}{\sqrt{1-\cos^2 x}}=\frac{-1}{9}\int\frac{5-4y}{3\sqrt{1-y^2}}dy=-\frac{1}{27}\int\frac{5-4y}{\sqrt{1-y^2}}dy$$

$$=-\frac{1}{27}\left(5\sin^{-1}y+4\sqrt{1-y^2}\right)=\frac{5}{27}\cos^{-1}y-\frac{4}{27}\sqrt{1-y^2}+c$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question I04/3/076 to I04/3/077

Given $f(x)=x\ln x$. Let Δ denotes the area bounded by $y=f(x), x=a, x=a^3$ and x-axis, for $0<a<1$.

$f(x)=x\ln x$ दिया गया है। माना Δ, $y=f(x), x=a, x=a^3$, $0<a<1$ तथा x-अक्ष से परिबद्ध क्षेत्रफल को दर्शाता है।

I04/3/076. Mark the correct statement :

(A) Δ is maximum when $a=1/\sqrt{2}$ (B) Δ is minimum when $a=1/2$

(C) Δ is maximum when $a=1/\sqrt{3}$ (D) None of these

सही कथन होगा :

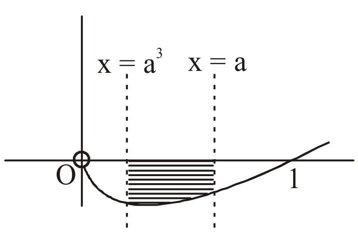
(A) Δ अधिकतम होगा जब $a=1/\sqrt{2}$ (B) Δ न्यूनतम होगा जब $a=1/2$

(C) Δ अधिकतम होगा जब $a=1/\sqrt{3}$ (D) इनमें से कोई नहीं

Ans. C

Sol. From figure,

$\Delta(a)=-\int_{a^3}^{a} x\ln x\ dx=\int_{a}^{a^3} x\ln x dx.$



$\Delta'(a)=a^3\ln(a^3)3a^2-a\ln a.1$ (using Leibniz Rule). So $\Delta'(a)=a\ln a(9a^4-1)=0$ gives $a=1/\sqrt{3}$.

$\Delta'(a)$ changes sign from positive to negative while passing through $1/\sqrt{3}$ so it is point of local maximum.

I04/3/077. Mark the correct statement :

(A) The maximum value of Δ is $(4 + \ln 27)/54$

(B) The maximum value of Δ is $(2 + \ln 9)/32$

(C) The minimum value of Δ is $(7 + \ln 18)/63$

(D) None of these

सत्य कथन होगा :

(A) Δ का अधिकतम मान $(4 + \ln 27)/54$ होगा

(B) Δ का अधिकतम मान $(2 + \ln 9)/32$ होगा

(C) Δ का न्यूनतम मान $(7 + \ln 18)/63$ होगा

(D) इनमें से कोई नहीं

Ans. A

Sol. $\Delta(a) = \int_a^{a^3} x \ln x dx$. Use integration by parts taking ln x as first function, to get $\Delta(a)$. Then put $a = 1/\sqrt{3}$.

Paragraph for Question I04/3/078 to I04/3/079

A curve $y = f(x)$ passes through the point P(1, 1) and the normal to the curve at P is $a(y-1) + (x-1) = 0$. If the slope of the tangent at any point on the curve is proportional to the ordinate of that point, then :

एक वक्र $y = f(x)$ एक बिन्दु P(1, 1) से गुजरता है तथा वक्र का बिन्दु P पर $a(y-1) + (x-1) = 0$ अभिलम्ब है। यदि वक्र के किसी बिन्दु पर स्पर्शरेखा का ढाल उस बिन्दु के y निर्देशांक के समानुपाती है, तब :

I04/3/078. The equation of the curve $y = f(x)$ is :

वक्र $y = f(x)$ का समीकरण होगा :

(A) $y = e^{a(x-1)}$ (B) $y = e^{a(x^2-1)}$ (C) $y = e^{a(1-|x|)}$ (D) $y = e^{a(|x|-1)}$

Ans. A

Sol. Slope of tangent at any point P (x, y) is $\frac{dy}{dx} \propto y$

$\Rightarrow \frac{dy}{dx} = ky \Rightarrow y = e^{k(x-1)}$

$\therefore a(y-1) + (x-1) = 0$ is normal at (1, 1)

$\Rightarrow k = a$

Hence $f(x) = e^{a(x-1)}$

I04/3/079. The area bounded by the curve $y = f(x)$, y-axis and the normal to the curve at P is :

वक्र $y = f(x)$, y-अक्ष तथा वक्र के बिन्दु P पर अभिलम्ब से परिबद्ध क्षेत्रफल होगा :

(A) $\frac{1}{a}\left(a-\frac{1}{2}+e^{-a}\right)$ (B) $\frac{1}{a}\left(a+\frac{1}{2}+e^{-a}\right)$ (C) $\frac{1}{a}\left(a-\frac{1}{2}+e^{a}\right)$ (D) $\frac{1}{a}\left(a+\frac{1}{2}+e^{a}\right)$

Ans. A

Sol. Area bounded $=\int_0^1\left(\frac{1}{a}-\frac{x}{a}+1-e^{a(x-1)}\right)dx$

$=\frac{1}{a}\left(a-\frac{1}{2}+e^{-a}\right)$ sq. unit

Paragraph for Question I04/3/080 to I04/3/082

For $i = 0,1,2,\ldots\ldots,n$, let S_i denotes the area of region bounded by the curve $y = e^{-2x}\sin x$ with x-axis from $x = i\pi$ to $x = (i+1)\pi$.

$i = 0,1,2,\ldots\ldots,n$ के लिये माना S_i वक्र $y = e^{-2x}\sin x$ का x-अक्ष के साथ क्षेत्रफल $x = i\pi$ से $x = (i+1)\pi$ के बीच है।

I04/3/080. The value of S_0 is :

S_0 का मान है :

(A) $\frac{1+e^{2\pi}}{5}$ (B) $\frac{1-e^{-2\pi}}{5}$ (C) $\frac{1+e^{-2\pi}}{5}$ (D) $\frac{1+e^{-\pi}}{5}$

Ans. C

Sol. Area $=\int_0^{\pi} e^{-2x}\sin x\,dx$

$=\left[\frac{e^{-2x}}{5}[-2\sin x-\cos x]\right]_0^{\pi}=\frac{1+e^{-2\pi}}{5}$

I04/3/081. The ratio $\frac{S_{2014}}{S_{2015}}$ is equal to :

अनुपात $\frac{S_{2014}}{S_{2015}}$ का मान है :

(A) $e^{-2\pi}$ (B) $e^{2\pi}$ (C) $2e^{\pi}$ (D) $e^{-\pi}$

Ans. B

Sol.

I04/3/082. The value of $\sum_{i=0}^{\infty} S_i$ is equal to :

$\sum_{i=0}^{\infty} S_i$ का मान है :

(A) $\frac{e^{\pi}\left(1+e^{\pi}\right)}{5\left(e^{\pi}-1\right)}$ (B) $\frac{e^{2\pi}\left(e^{2\pi}+1\right)}{5\left(e^{2\pi}-1\right)}$ (C) $\frac{e^{2\pi}+1}{5\left(e^{2\pi}-1\right)}$ (D) $\frac{e^{2\pi}+1}{e^{2\pi}-1}$

Ans. C

Sol. $S_i = \int_{i\pi}^{(i+1)\pi} e^{-2x} \sin x dx = \left[\frac{e^{-2x}}{5}[-2\sin x - \cos x]\right]_{i\pi}^{(i+1)\pi}$

Paragraph for Question I04/3/083 to I04/3/0085

Let $f : A \to B$ $f(x) = \frac{x+a}{bx^2+cx+2}$, where A represent domain set and B represent range set of function f(x), a, b, c $\in$ R, f(–1) = 0 and y = 1 is an asymptote of y = f(x) and y = g(x) is the inverse of f(x).

माना $f : A \to B$ $f(x) = \frac{x+a}{bx^2+cx+2}$, जहाँ A प्रांत समुच्चय तथा B फलन f(x), a, b, c $\in$ R, f(–1) = 0 के परिसर समुच्चय को दर्शाता है तथा y = 1, y = f(x) का एक अनन्तस्पर्शी है तथा y = g(x), f(x) का व्युत्क्रम है।

I04/3/083. g(0) is equal to :

g(0) बराबर है :

(A) –1 (B) –3 (C) $-\frac{5}{2}$ (D) $-\frac{3}{2}$

Ans. A

Sol.

I04/3/084. Area bounded between the curves y = f(x) and y = g(x) is :

वक्रों y = f(x) तथा y = g(x) से परिबद्ध क्षेत्रफल होगा :

(A) $2\sqrt{5} + \ell n\left(\frac{3-\sqrt{5}}{3+\sqrt{5}}\right)$ (B) $3\sqrt{5}+2\ell n\left(\frac{3+\sqrt{5}}{3-\sqrt{5}}\right)$

(C) $3\sqrt{5}+4\ell n\left(\frac{3-\sqrt{5}}{3+\sqrt{5}}\right)$ (D) $3\sqrt{5}+2\ell n\left(\frac{3-\sqrt{5}}{3+\sqrt{5}}\right)$

Ans. D

Sol.

I04/3/085. Area of region enclosed by asymptotes of curves $y = f(x)$ and $y = g(x)$ is :

वक्रों $y = f(x)$ तथा $y = g(x)$ के अनन्तस्पर्शीयों द्वारा घिरा हुआ क्षेत्रफल होगा :

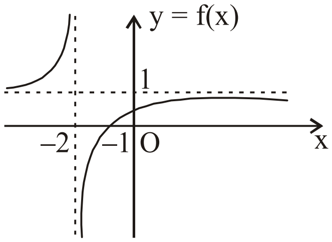
(A) 4 (B) 9 (C) 12 (D) 25

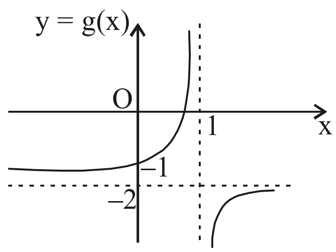
Ans. B

Sol. $f(x) = \dfrac{x+a}{bx^2+cx+2}$

$f(-1) = 0 \Rightarrow a = 1$

If $y = 1$ is asymptotes, then $b = 0$ and $c = 1 \Rightarrow f(x) = \dfrac{x+1}{x+2}$ and $g(x) = \dfrac{1-2x}{x-1}$





Paragraph for Question I04/3/086 to I04/3/088

For $j = 0, 1, 2, ...\ n$ let S_j be the area of region bounded by the x-axis and the curve $ye^x = \sin x$ for $j\pi \le x \le (j+1)\pi$

$j = 0, 1, 2, ...\ n$ के लिए माना S_j, x-axis तथा वक्र $ye^x = \sin x$ से परिबद्ध क्षेत्रफल है जहाँ $j\pi \le x \le (j+1)\pi$

I04/3/086. The value of S_0 is :

S_0 का मान है :

(A) $\frac{1}{2}(1+e^{\pi})$ (B) $\frac{1}{2}(1+e^{-\pi})$ (C) $\frac{1}{2}(1-e^{-\pi})$ (D) $\frac{1}{2}(e^{\pi}-1)$

Ans. B

Sol.

I04/3/087. The ratio $\dfrac{S_{2009}}{S_{2010}}$ equals :

$\frac{S_{2009}}{S_{2010}}$ बराबर होगा :

(A) $e^{-\pi}$ (B) e^{π} (C) $\frac{1}{2}e^{\pi}$ (D) $2e^{\pi}$

Ans. B

Sol.

I04/3/088. The value of $\sum_{j=0}^{\infty} S_j$ equals to :

$\sum_{j=0}^{\infty} S_j$ का मान होगा :

(A) $\frac{e^{\pi}(1+e^{\pi})}{2(e^{\pi}-1)}$ (B) $\frac{1+e^{\pi}}{2(e^{\pi}-1)}$ (C) $\frac{1+e^{\pi}}{e^{\pi}-1}$ (D) $\frac{e^{\pi}(1+e^{\pi})}{(e^{\pi}-1)}$

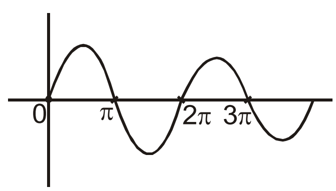
Ans. B

Sol. $y = e^{-x}\sin x$

$\frac{dy}{dx} = e^{-x}[\cos x - \sin x] = 0 \Rightarrow \tan x = 1 \Rightarrow x = n\pi + \pi/4$

So (i.e. $0 \le x \le \pi$)

$I = \int e^{-x}\sin x\,dx$



$I = \frac{-e^{-x}}{2}[\sin x + \cos x] + c$

$$s_j = \left|\int_{j\pi}^{(j+1)\pi} e^{-x}\sin x\,dx\right| = \left|-\frac{e^{-x}}{2}[\sin x + \cos x]\Big|_{j\pi}^{(j+1)\pi}\right| = \frac{e^{-j\pi}}{2}\left(e^{-\pi}+1\right)$$

86. Put $j = 0$, $s_0 = \frac{1+e^{-\pi}}{2}$

87. $\frac{s_{2009}}{s_{2010}} = \frac{e^{-2009\pi}}{e^{-2010\pi}} = e^{\pi}$

88. $\frac{s_{j+1}}{s_j} = e^{-\pi}$ $\because \sum_{j=0}^{\infty} S_j = \frac{s_0}{1-e^{-\pi}} = \frac{\frac{1+e^{-\pi}}{2}}{1-e^{-\pi}} = \frac{1+e^{\pi}}{2(e^{\pi}-1)}$

Paragraph for Question I02/3/089 to I02/3/090

Let $I_1 = \int_0^1 (1-x^2)^{1/3} dx$ & $I_2 = \int_0^1 (1-x^3)^{1/2} dx$.

On the basis of above information, answer the following questions :

माना $I_1 = \int_0^1 (1-x^2)^{1/3} dx$ तथा $I_2 = \int_0^1 (1-x^3)^{1/2} dx$ है।

उपरोक्त जानकारी के आधार पर, निम्न प्रश्नों के उत्तर दीजिए :

I02/3/089. I_1 is equal to :

I_1 का मान होगा :

(A) $\frac{2}{3}\int_0^{\pi/2} (\sin^2\theta)(\cos\theta)^{-1/3} d\theta$

(B) $\frac{3}{2}\int_0^{\pi/2} (\sin^2\theta)(\cos\theta)^{-1/3} d\theta$

(C) $\frac{2}{3}\int_0^{\pi/2} (\sin\theta)^{2/3}(\cos\theta)^{-1/3} d\theta$

(D) $\frac{3}{2}\int_0^{\pi/2} (\sin\theta)^{2/3}(\cos\theta)^{-1/3} d\theta$

Ans. C

Sol. $I_1 = \int_0^1 (1-x^2)^{1/3} dx = (1-x^2))_0^3 + \frac{2}{3}\int_0^1 \frac{x^2}{(1-x^2)^{2/3}} dx$

$x = \sin\theta, dx = \cos\theta\, d\theta$

$$I_1 = \frac{2}{3}\int_0^{\pi/2} \frac{\sin^2\theta\cos\theta}{(\cos\theta)^{4/3}} d\theta$$

$$I_1 = \frac{2}{3}\int_0^{\pi/2} \sin^2\theta(\cos\theta)^{-1/3} d\theta$$

I02/3/090. I_2 is equal to :

I_2 का मान होगा :

(A) $\frac{2}{3}\int_0^{\pi/2} \frac{\sin^3\theta\cos\theta}{\sqrt{1-\sin^3\theta}} d\theta$

(B) $\frac{2}{3}\int_0^{\pi/2} \frac{\cos^3\theta\sin\theta}{\sqrt{1-\sin^3\theta}}$

(C) $\frac{3}{2}\int_0^{\pi/2} \frac{\sin^3\theta\cos\theta}{\sqrt{1-\sin^3\theta}} d\theta$

(D) $\frac{2}{3}\int_0^{\pi/2} \frac{\sin^3\theta\cos\theta}{\sqrt{1-\sin^3\theta}} d\theta$

Ans. C

Sol. $I_2 = \int_0^1 (1-x^3)^{1/2}\,dx$

$= x(1-x^3)^{1/2}\Big)_0^1 + \frac{3}{2}\int_0^1 \frac{x^3 dx}{\sqrt{1-x^3}}$

Put $x = \sin\theta$

$I_2 = \frac{3}{2}\int_0^{\pi/2} \frac{\sin^3\theta\cos\theta}{\sqrt{1-\sin^3\theta}}\,d\theta$

Paragraph for Question I02/3/091 to I02/3/092

Let $f(x) = e^{\sin x}$ and $g(2-x) = x^2 - f(x)$.

On the basis of above information, answer the following questions :

माना $f(x) = e^{\sin x}$ तथा $g(2-x) = x^2 - f(x)$ है।

उपरोक्त जानकारी के आधार पर, निम्न प्रश्नों के उत्तर दीजिए :

I02/3/091. $\int_0^{\frac{32\pi}{3}} \left|\ell n(f(x))\right| dx$ is :

$\int_0^{\frac{32\pi}{3}} \left|\ell n(f(x))\right| dx$ का मान होगा :

(A) $\frac{41}{2}$ (B) $\frac{43}{2}$ (C) $\frac{45}{2}$ (D) $\frac{47}{2}$

Ans. B

Sol. $\int_0^{\frac{32\pi}{3}} |\sin x|dx \Rightarrow \int_0^{10\pi} |\sin x|\,dx + \int_0^{\frac{2\pi}{3}} |\sin x|\,dx$

$\Rightarrow \quad 10\int_0^{\pi} \sin x dx + \int_0^{2\pi/3} \sin x dx$

$= 20 + \frac{3}{2} = \frac{43}{2}$

I02/3/092. If $\int_0^2 \left(f(x)+g(x)\right)dx = k$ then [k] is (where [.] denotes greatest integer function) :

यदि $\int_0^2 \left(f(x)+g(x)\right)dx = k$ हो, तो [k] का मान होगा (जहाँ [.] महत्तम पूर्णांक फलन को दर्शाता है) :

(A) 0 (B) 2 (C) 4 (D) 5

Ans. B

Sol. $\int_0^2 f(x)dx + \int_0^2 g(x)dx$

$\int_0^2 f(x)dx + \int_0^2 g(2-x)dx \Rightarrow \int_0^2 x^2dx = \frac{8}{3}$

$\left[\frac{8}{3}\right] = 2$

Paragraph for Question I02/3/093 to I02/3/095

Let $f(x)$ is a derivable function satisfying $xf(x) - \int_0^x f(t)dt = x + \ln\left(\sqrt{x^2+1}-x\right)$ with $f(0) = \ln 2$.

Let $g(x) = x\,f'(x)$ then

माना $f(x)$ एक अवकलनीय फलन है जो इस प्रकार है कि $xf(x) - \int_0^x f(t)dt = x + \ln\left(\sqrt{x^2+1}-x\right)$ जहाँ $f(0) = \ln 2$.

माना $g(x) = x\,f'(x)$ तो

I02/3/093. Range of g(x) is :

फलन g(x) का परिसर है :

(A) $[0,\infty)$ (B) $[0,1)$ (C) $[1,\infty)$ (D) $(-\infty,\infty)$

Ans. B

Sol. $xf(x) - \int_0^x f(t)dt = x + \ell n^{\left(\sqrt{1+x^2}-x\right)}$,

$f(0) = ln2$

Differentiating with respect to x

$$f(x) + xf'(x) - f(x) = 1 + \left(\frac{1}{\sqrt{1+x^2} - x}\right)\left(\frac{x}{\sqrt{1+x^2}} - 1\right)$$

$$xf'(x) = 1 + \left(\frac{1}{\sqrt{1+x^2} - x}\right)\left(\frac{x - \sqrt{1+x^2}}{\sqrt{1+x^2}}\right)$$

$$xf'(x) = 1 - \frac{1}{\sqrt{1+x^2}}$$

$\therefore\ g(x) = xf'(x)$

$$\Rightarrow g(x) = 1 - \frac{1}{\sqrt{1+x^2}}$$

$\because\ 1 + x^2 \in [1, \infty)$

$\sqrt{1+x^2} \in [1, \infty)$

$$1 - \frac{1}{\sqrt{1+x^2}} \in (0, 1]$$

$g(x) \in [0, 1)$

I02/3/094. For the function f which one of the following is correct ?

(A) f is neither odd nor even
(B) f is transcendental
(C) f is injective
(D) f is symmetric w.r.t. origin

फलन f के लिए निम्न में कौनस सत्य है ?

(A) f ना तो विषम है और ना ही सम फलन है
(B) f परिवर्तनीय फलन है
(C) f ऐकाकी है
(D) f मूल बिन्दु के सापेक्ष सममित है

Ans. B

Sol. $\because\ xf'(x) = 1 - \dfrac{1}{\sqrt{1+x^2}}$

$$= \frac{\sqrt{1+x^2} - 1}{\sqrt{1+x^2}} \times \left(\frac{\sqrt{1+x^2} + 1}{\sqrt{1+x^2} + 1}\right)$$

$$xf'(x) = \frac{x^2}{\left(\sqrt{1+x^2}\right)\left(\sqrt{1+x^2} + 1\right)}$$

$$f'(x)=\frac{x}{\left(\sqrt{1+x^2}\right)\left(\sqrt{1+x^2}+1\right)}$$

Integrating w.r.t. x

$$f(x)=\int\frac{x}{\left(\sqrt{1+x^2}\right)\left(\sqrt{1+x^2+1}\right)}$$

Let $1+x^2=t^2$

$$2xdx=2tdt$$
$$xdx=tdt$$

$$f(x)=\ell n|t+1|+c$$

$$f(x)=\ell n\left|\sqrt{1+x^2}+1\right|+c$$

$$\because f(0)=\ell n2$$

$$\ell n2=\ell n2+c \quad \Rightarrow c=0$$

$\Rightarrow f(x)=\ell n\left(\sqrt{1+x^2}+1\right)$ $\quad \Rightarrow$ f(x) is transcedental function

f(x) = f(–x) $\Rightarrow$ f(x) is even function

$\Rightarrow$ symmetric about y-axis

$$\therefore f'(x)=\frac{1}{\sqrt{1+x^2}+1}\times\frac{2x}{2\sqrt{1+x^2}}=\frac{x}{\left(\sqrt{1+x^2}\right)\left(\sqrt{1+x^2}+1\right)}$$

I02/3/095. $\int_0^1 f(x)dx=:$

(A) $ln\left(3+2\sqrt{2}\right)-1$ (B) $2ln\left(1+\sqrt{2}\right)$ (C) $ln\left(1+\sqrt{2}\right)-1$ (D) 1

Ans. A

Sol. $$\int_0^1 f(x)dx=\int_0^1 \ell n\left(1+\sqrt{1+x^2}\right)dx$$

$$=\left[x\ell n\left(1+\sqrt{1+x^2}\right)\right]_0^1-\int_0^1\left(\frac{1}{1+\sqrt{1+x^2}}\times\frac{2x}{2\sqrt{1+x^2}}\times x\right)dx$$

$$\ell n\left(1+\sqrt{2}\right)-\int_0^1\frac{x^2\left[1-\sqrt{1+x^2}\right]}{\left[\sqrt{1+x^2}\left(-x^2\right)\right]}dx$$

$$= \ell n\left(1+\sqrt{2}\right)+\int_0^1\left(\frac{1}{\sqrt{1+x^2}}-1\right)dx$$

$$= \ell n\left(1+\sqrt{2}\right)+\left[\ell n x+\sqrt{1+x^2}\right]_0^1-\left[x\right]_0^1$$

$$\ell n\left(1+\sqrt{2}\right)+\left[\ell n\left(1+\sqrt{2}\right)-\ell n 1\right]-1$$

$$= 2\ell n\left(1+\sqrt{2}\right)-1$$

$$= \ell n\left(1+\sqrt{2}\right)^2-1$$

$$= \ell n\left(3+2\sqrt{2}\right)-1$$

Paragraph for Question I02/3/096 to I02/3/098

Consider the functions $f : R \to R$ & $g : R \to R$ such that $f(x) = (\sin x + \cos x) - x\int_0^x g(t)dt$ &

$g(x) = x^2 - \int_0^1 f(t)dt = x^2 - c$.

On the basis of above information, answer the following questions

माना कि फलन $f : R \to R$ तथा $g : R \to R$ इस प्रकार है कि $f(x) = (\sin x + \cos x) - x\int_0^x g(t)dt$ तथा

$g(x) = x^2 - \int_0^1 f(t)dt = x^2 - c$.

उपरोक्त जानकारी के आधार पर, निम्न प्रश्नों के उतर दीजिए।

I02/3/096. The value of $\lim_{x\to 0}\frac{xf(x)}{g'(x)}$ is :

$\lim_{x\to 0}\frac{xf(x)}{g'(x)}$ का मान है :

(A) $\frac{1}{2}$ (B) 1 (C) $\frac{3}{2}$ (D) 2

Ans. A

Sol. $f(x) = \sin x + \cos x - x\int_0^x \left(x^2 - c\right)$

$\sin x + \cos x - x\left[\frac{x^3}{3} - cx\right]_0^x$

$f(x) = \sin x + \cos x + c^2 - \frac{x^4}{3}$

$c = \int_0^1 \left(\sin x + \cos x + \cos^2 - \frac{x^4}{3}\right)\cdot dx$

$= \left(-\cos x + \sin x + \frac{cx^3}{3} - \frac{x^5}{15}\right)_0^1$

$c = 1 - \cos 1 + \sin 1 + \frac{c}{3} - \frac{1}{15}$

$\frac{2c}{3} = \frac{14}{15} + \sin 1 - \cos 1$

$c = \frac{3}{2}\left(\frac{14}{15} + \sin 1 - \cos 1\right)$

$L = \frac{x\left(\sin x + \cos x + cx^2 - \frac{x^4}{3}\right)}{2x}$

$= \frac{1}{2}$

I02/3/097. The value of $\int\limits_{-1}^{1} (f(x) + g(x))\,dx$ is :

(A) a positive number

(B) a negative number

(C) 0

(D) $\frac{1}{2}$

$\int\limits_{-1}^{1} (f(x) + g(x))\,dx$ का मान है :

(A) एक धनात्मक संख्या

(B) एक ऋणात्मक संख्या

(C) 0

(D) $\frac{1}{2}$

Ans. B

Sol. $$\int_{-1}^{1}\left(f(x)+g(x)\right)dx=\int\left[\left(\sin x+\cos x-\frac{x^4}{3}+c^2\right)-\left(x^2-c\right)\right]dx$$

$$=\int_{-1}^{1}\left(\sin x+c^2+c\right)dx+\int_{-1}^{1}\left(\cos x-\frac{x^4}{3}-x^2\right)dx$$

$$=0+2\int_{0}^{1}\left(\cos x-\frac{x^4}{3}-x^2\right)dx$$

$$=2\left[\sin x-\frac{x^5}{15}-\frac{x^3}{3}\right]_0^1$$

$$=2\left[\sin 1-\frac{1}{15}-\frac{1}{3}\right]$$

$$=2\left[\sin 1-\frac{8}{15}\right]<0$$

= negative number

I02/3/098. Let $h(x)=f(x)+x^2\,g(x)$, then choose the correct option :

(A) h(x) has maxima but not minima
(B) h(x) has minima but not maxima
(C) h(x) has exactly one maxima & one minima
(D) The range of h(x) is R

यदि $h(x)=f(x)+x^2\,g(x)$ है, तो सही कथन चुनिये :

(A) h(x) का उच्चिष्ठ मान है किन्तु निम्निष्ठ मान नहीं
(B) h(x) का निम्निष्ठ मान है किन्तु उच्चिष्ठ मान नहीं
(C) h(x) केवल एक उच्चिष्ठ और एक निम्निष्ठ मान रखता है
(D) h(x) का परिसर R है

Ans. B

Sol. $$2\left[\sin x+\frac{(c+1)x^5}{3}-\frac{x^5}{15}-cx\right]_0^1$$

$$2\left[\sin 1+\frac{c}{3}+\frac{1}{3}-c-\frac{1}{15}\right]$$

$$2\left[\sin 1-\frac{2c}{3}+\frac{4}{15}\right]$$